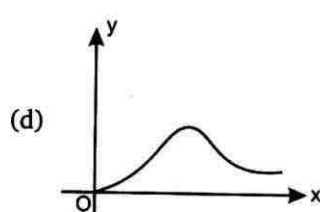
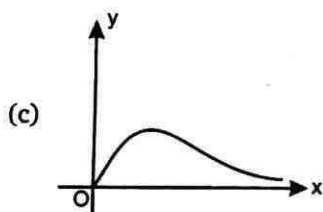
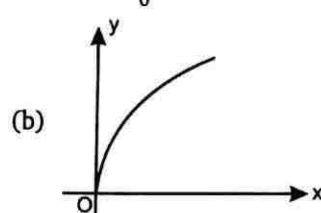
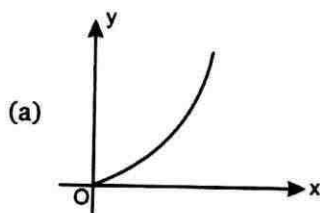


35. If $f(x) = 2x$, $g(x) = 3 \sin x - x \cos x$, then for $x \in \left(0, \frac{\pi}{2}\right)$:
- (a) $f(x) > g(x)$ (b) $f(x) < g(x)$
 (c) $f(x) = g(x)$ has exactly one real root. (d) $f(x) = g(x)$ has exactly two real roots
36. Let $f(x) = \sin^{-1} \left(\frac{2g(x)}{1+g(x)^2} \right)$, then which are correct ?
- (i) $f(x)$ is decreasing if $g(x)$ is increasing and $|g(x)| > 1$
 (ii) $f(x)$ is an increasing function if $g(x)$ is increasing and $|g(x)| \leq 1$
 (iii) $f(x)$ is decreasing function if $g(x)$ is decreasing and $|g(x)| > 1$
 (a) (i) and (iii) (b) (i) and (ii) (c) (i), (ii) and (iii) (d) (iii)
37. The graph of the function $y = f(x)$ has a unique tangent at $(e^a, 0)$ through which the graph passes then $\lim_{x \rightarrow e^a} \frac{\ln(1+7f(x)) - \sin(f(x))}{3f(x)}$ is equal to :
- (a) 1 (b) 3 (c) 2 (d) 7
38. Let $f(x)$ be a function such that $f'(x) = \log_{1/3}(\log_3(\sin x + a))$. The complete set of values of 'a' for which $f(x)$ is strictly decreasing for all real values of x is :
- (a) $[4, \infty)$ (b) $[3, 4]$ (c) $(-\infty, 4)$ (d) $[2, \infty)$
39. If $f(x) = a \ln|x| + bx^2 + x$ has extremas at $x = 1$ and $x = 3$, then :
- (a) $a = \frac{3}{4}, b = -\frac{1}{8}$ (b) $a = \frac{3}{4}, b = \frac{1}{8}$ (c) $a = -\frac{3}{4}, b = -\frac{1}{8}$ (d) $a = -\frac{3}{4}, b = \frac{1}{8}$
40. Let $f(x) = \begin{cases} 1 + \sin x, & x < 0 \\ x^2 - x + 1, & x \geq 0 \end{cases}$, then :
- (a) f has a local maximum at $x = 0$ (b) f has a local minimum at $x = 0$
 (c) f is increasing everywhere (d) f is decreasing everywhere
41. If m and n are positive integers and $f(x) = \int_1^x (t-a)^{2n}(t-b)^{2m+1} dt$, $a \neq b$, then :
- (a) $x = b$ is a point of local minimum (b) $x = b$ is a point of local maximum
 (c) $x = a$ is a point of local minimum (d) $x = a$ is a point of local maximum
42. For any real θ , the maximum value of $\cos^2(\cos \theta) + \sin^2(\sin \theta)$ is :
- (a) 1 (b) $1 + \sin^2 1$
 (c) $1 + \cos^2 1$ (d) Does not exist
43. If the tangent at P of the curve $y^2 = x^3$ intersects the curve again at Q and the straight line OP, OQ have inclinations α, β where O is origin, then $\left(\frac{\tan \alpha}{\tan \beta} \right)$ has the value, equals to :
- (a) -1 (b) -2 (c) 2 (d) $\sqrt{2}$

44. If $x + 4y = 14$ is a normal to the curve $y^2 = \alpha x^3 - \beta$ at $(2, 3)$, then value of $\alpha + \beta$ is :
 (a) 9 (b) -5 (c) 7 (d) -7
45. The tangent to the curve $y = e^{kx}$ at a point $(0, 1)$ meets the x -axis at $(a, 0)$ where $a \in [-2, -1]$, then $k \in$:
 (a) $\left[-\frac{1}{2}, 0\right]$ (b) $\left[-1, -\frac{1}{2}\right]$ (c) $[0, 1]$ (d) $\left[\frac{1}{2}, 1\right]$

46. Which of the following graph represent the function $f(x) = \int_0^{\sqrt{x}} e^{-\frac{u^2}{x}} du$, for $x > 0$ and $f(0) = 0$?



47. Let $f(x) = (x-a)(x-b)(x-c)$ be a real valued function where $a < b < c$ ($a, b, c \in \mathbb{R}$) such that $f''(\alpha) = 0$. Then if $\alpha \in (c_1, c_2)$, which one of the following is correct ?
 (a) $a < c_1 < b$ and $b < c_2 < c$ (b) $a < c_1, c_2 < b$
 (c) $b < c_1, c_2 < c$ (d) None of these
48. $f(x) = x^6 - x - 1$, $x \in [1, 2]$. Consider the following statements :
 (1) f is increasing on $[1, 2]$ (2) f has a root in $[1, 2]$
 (3) f is decreasing on $[1, 2]$ (4) f has no root in $[1, 2]$
 Which of the above are correct?
 (a) 1 and 2 (b) 1 and 4 (c) 2 and 3 (d) 3 and 4
49. Which one of the following curves is the orthogonal trajectory of straight lines passing through a fixed point (a, b) ?
 (a) $x - a = k(y - b)$ (b) $(x - a)(y - b) = k$
 (c) $(x - a)^2 = k(y - b)$ (d) $(x - a)^2 + (y - b)^2 = k$

50. The function $f(x) = \sin^3 x - m \sin x$ is defined on open interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ and if assumes only 1 maximum value and only 1 minimum value on this interval. Then, which one of the following must be correct ?
 (a) $0 < m < 3$ (b) $-3 < m < 0$ (c) $m > 3$ (d) $m < -3$
51. The greatest of the numbers $1, 2^{1/2}, 3^{1/3}, 4^{1/4}, 5^{1/5}, 6^{1/6}$ and $7^{1/7}$ is :
 (a) $2^{1/2}$ (b) $3^{1/3}$ (c) $7^{1/7}$ (d) $6^{1/6}$
52. Let l be the line through $(0, 0)$ and tangent to the curve $y = x^3 + x + 16$. Then the slope of l equal to :
 (a) 10 (b) 11 (c) 17 (d) 13
53. The slope of the tangent at the point of inflection of $y = x^3 - 3x^2 + 6x + 2009$ is equal to :
 (a) 2 (b) 3 (c) 1 (d) 4
54. Let f be a real valued function with $(n+1)$ derivatives at each point of R . For each pair of real numbers $a, b, a < b$, such that

$$\ln \left[\frac{f(b) + f'(b) + \dots + f^{(n)}(b)}{f(a) + f'(a) + \dots + f^{(n)}(a)} \right] = b - a$$

Statement-1 : There is a number $c \in (a, b)$ for which $f^{(n+1)}(c) = f(c)$

because

Statement-2 : If $h(x)$ be a derivable function such that $h(p) = h(q)$ then by Rolle's theorem $h'(d) = 0, d \in (p, q)$

- (a) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1
 (b) Statement-1 is true, statement-2 is true and statement-2 is not correct explanation for statement-1
 (c) Statement-1 is true, statement-2 is false
 (d) Statement-1 is false, statement-2 is true
55. If $g(x)$ is twice differentiable real valued function satisfying $g''(x) - 3g'(x) > 3 \forall x \geq 0$ and $g'(0) = -1$, then $h(x) = g(x) + x \forall x > 0$ is :
 (a) strictly increasing (b) strictly decreasing
 (c) non monotonic (d) data insufficient
56. If the straight line joining the points $(0, 3)$ and $(5, -2)$ is tangent to the curve $y = \frac{c}{x+1}$; then the value of c is :
 (a) 2 (b) 3 (c) 4 (d) 5
57. Number of solutions(s) of $\ln |\sin x| = -x^2$ if $x \in \left[-\frac{\pi}{2}, \frac{3\pi}{2}\right]$ is/are :
 (a) 2 (b) 4 (c) 6 (d) 8

58. The equation $\sin^{-1} x = |x - a|$ will have atleast one solution then complete set of values of a be :
- (a) $[-1, 1]$ (b) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (c) $\left[1 - \frac{\pi}{2}, 1 + \frac{\pi}{2}\right]$ (d) $\left[\frac{\pi}{2} - 1, \frac{\pi}{2} + 1\right]$
59. For any real number b , let $f(b)$ denotes the maximum of $\left|\sin x + \frac{2}{3 + \sin x} + b\right| \forall x \in R$.
- Then the minimum value of $f(b) \forall b \in R$ is :
- (a) $\frac{1}{2}$ (b) $\frac{3}{4}$ (c) $\frac{1}{4}$ (d) 1
60. Which of the following are correct
- (a) $x^4 + 2x^2 - 6x + 2 = 0$ has exactly four real solution
- (b) $x^5 + 5x + 1 = 0$ has exactly three real solutions
- (c) $x^n + ax + b = 0$ where n is an even natural number has atmost two real solution $a, b \in R$.
- (d) $x^3 - 3x + c = 0, c > 0$ has two real solution for $x \in (0, 1)$
61. For any real number b , let $f(b)$ denotes the maximum of $\left|\sin x + \frac{2}{3 + \sin x} + b\right| \forall x \in R$. Then the minimum value of $f(b) \forall b \in R$ is :
- (a) $\frac{1}{2}$ (b) $\frac{3}{4}$ (c) $\frac{1}{4}$ (d) 1
62. If p be a point on the graph of $y = \frac{x}{1 + x^2}$, then coordinates of 'p' such that tangent drawn to curve at p has the greatest slope in magnitude is :
- (a) $(0, 0)$ (b) $\left(\sqrt{3}, \frac{\sqrt{3}}{4}\right)$ (c) $\left(-\sqrt{3}, -\frac{\sqrt{3}}{4}\right)$ (d) $\left(1, \frac{1}{2}\right)$
63. Let $f: [0, 2\pi] \rightarrow [-3, 3]$ be a given function defined as $f(x) = 3 \cos \frac{x}{2}$. The slope of the tangent to the curve $y = f^{-1}(x)$ at the point where the curve crosses the y -axis is :
- (a) -1 (b) $-\frac{2}{3}$ (c) $-\frac{1}{6}$ (d) $-\frac{1}{3}$
64. Number of stationary points in $[0, \pi]$ for the function $f(x) = \sin x + \tan x - 2x$ is :
- (a) 0 (b) 1 (c) 2 (d) 3
65. If $a, b, c, d \in R$ such that $\frac{a + 2c}{b + 3d} + \frac{4}{3} = 0$, then the equation $ax^3 + bx^2 + cx + d = 0$ has
- (a) atleast one root in $(-1, 0)$ (b) atleast one root in $(0, 1)$
- (c) no root in $(-1, 1)$ (d) no root in $(0, 2)$

66. If $f'(x) = \phi(x)(x-2)^2$. Where $\phi(2) \neq 0$ and $\phi(x)$ is continuous at $x=2$, then in the neighbourhood of $x=2$
- (a) f is increasing if $\phi(2) < 0$ (b) f is decreasing if $\phi(2) > 0$
 (c) f is neither increasing nor decreasing (d) f is increasing if $\phi(2) > 0$
67. If $f(x) = x^3 - 6x^2 + ax + b$ is defined on $[1, 3]$ satisfies Rolle's theorem for $c = \frac{2\sqrt{3}+1}{\sqrt{3}}$ then
- (a) $a = -11, b = 6$ (b) $a = -11, b = -6$ (c) $a = 11, b \in R$ (d) $a = 22, b = -6$
68. For which of the following function(s) Lagrange's mean value theorem is not applicable in $[1, 2]$?
- (a) $f(x) = \begin{cases} \frac{3}{2} - x, & x < \frac{3}{2} \\ \left(\frac{3}{2} - x\right)^2, & x \geq \frac{3}{2} \end{cases}$ (b) $f(x) = \begin{cases} \frac{\sin(x-1)}{x-1}, & x \neq 1 \\ 1, & x = 1 \end{cases}$
 (c) $f(x) = (x-1)|x-1|$ (d) $f(x) = |x-1|$
69. If the curves $\frac{x^2}{a^2} + \frac{y^2}{4} = 1$ and $y^2 = 16x$ intersect at right angles, then :
- (a) $a = \pm 1$ (b) $a = \pm \sqrt{3}$ (c) $a = \pm \frac{1}{\sqrt{3}}$ (d) $a = \pm \sqrt{2}$
70. If the line $x \cos \alpha + y \sin \alpha = P$ touches the curve $4x^3 = 27ay^2$, then $\frac{P}{a} =$
- (a) $\cot^2 \alpha \cos \alpha$ (b) $\cot^2 \alpha \sin \alpha$ (c) $\tan^2 \alpha \cos \alpha$ (d) $\tan^2 \alpha \sin \alpha$

Answers

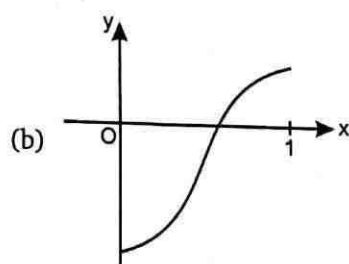
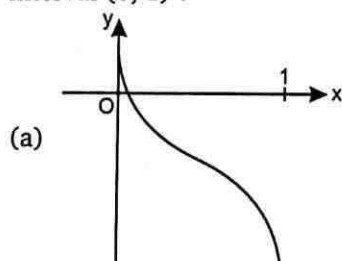
1. (d)	2. (b)	3. (a)	4. (c)	5. (b)	6. (c)	7. (d)	8. (d)	9. (c)	10. (b)
11. (c)	12. (c)	13. (c)	14. (c)	15. (c)	16. (a)	17. (b)	18. (a)	19. (b)	20. (b)
21. (b)	22. (d)	23. (d)	24. (a)	25. (d)	26. (c)	27. (b)	28. (d)	29. (c)	30. (d)
31. (b)	32. (c)	33. (d)	34. (d)	35. (a)	36. (b)	37. (c)	38. (a)	39. (c)	40. (a)
41. (a)	42. (b)	43. (b)	44. (a)	45. (d)	46. (b)	47. (a)	48. (a)	49. (d)	50. (a)
51. (b)	52. (d)	53. (b)	54. (a)	55. (a)	56. (c)	57. (b)	58. (c)	59. (b)	60. (c)
61. (b)	62. (a)	63. (b)	64. (c)	65. (b)	66. (d)	67. (c)	68. (a)	69. (d)	70. (a)

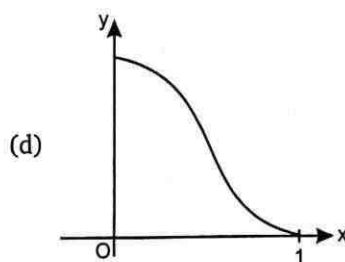
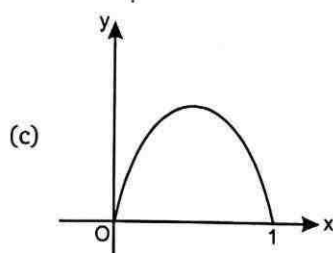
Exercise-2 : One or More than One Answer is/are Correct

- Common tangent(s) to $y = x^3$ and $x = y^3$ is/are :
 - $x - y = \frac{1}{\sqrt{3}}$
 - $x - y = -\frac{1}{\sqrt{3}}$
 - $x - y = \frac{2}{3\sqrt{3}}$
 - $x - y = \frac{-2}{3\sqrt{3}}$
- Let $f: [0, 8] \rightarrow \mathbb{R}$ be differentiable function such that $f(0) = 0$, $f(4) = 1$, $f(8) = 1$, then which of the following hold(s) good ?
 - There exist some $c_1 \in (0, 8)$ where $f'(c_1) = \frac{1}{4}$
 - There exist some $c \in (0, 8)$ where $f'(c) = \frac{1}{12}$
 - There exist $c_1, c_2 \in [0, 8]$ where $8f'(c_1)f(c_2) = 1$
 - There exist some $\alpha, \beta \in (0, 2)$ such that $\int_0^8 f(t) dt = 3(\alpha^2 f(\alpha^3) + \beta^2 f(\beta^3))$
- If $f(x) = \begin{cases} \sin^{-1}(\sin x) & x > 0 \\ \frac{\pi}{2} & x = 0 \\ \cos^{-1}(\cos x) & x < 0 \end{cases}$, then
 - $x = 0$ is a point of maxima
 - $f(x)$ is continuous $\forall x \in \mathbb{R}$
 - global maximum value of $f(x) \forall x \in \mathbb{R}$ is π
 - global minimum value of $f(x) \forall x \in \mathbb{R}$ is 0
- A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = \begin{cases} x^4 \left(2 + \sin \frac{1}{x} \right) & x \neq 0 \\ 0 & x = 0 \end{cases}$, then
 - f has a continuous derivative $\forall x \in \mathbb{R}$
 - f is a bounded function
 - f has an global minimum at $x = 0$
 - f'' is continuous $\forall x \in \mathbb{R}$
- If $|f''(x)| \leq 1 \forall x \in \mathbb{R}$, and $f(0) = 0 = f'(0)$, then which of the following can not be true ?
 - $f\left(-\frac{1}{2}\right) = \frac{1}{6}$
 - $f(2) = -4$
 - $f(-2) = 3$
 - $f\left(\frac{1}{2}\right) = \frac{1}{5}$
- Let $f: [-3, 4] \rightarrow \mathbb{R}$ such that $f''(x) > 0$ for all $x \in [-3, 4]$, then which of the following are always true ?
 - $f(x)$ has a relative minimum on $(-3, 4)$
 - $f(x)$ has a minimum on $[-3, 4]$
 - $f(x)$ has a maximum on $[-3, 4]$
 - if $f(3) = f(4)$, then $f(x)$ has a critical point on $[-3, 4]$

7. Let $f(x)$ be twice differentiable function such that $f''(x) > 0$ in $[0, 2]$. Then :
- $f(0) + f(2) = 2f(1)$, for atleast one $c, c \in (0, 2)$
 - $f(0) + f(2) < 2f(1)$
 - $f(0) + f(2) > 2f(1)$
 - $2f(0) + f(2) > 3f\left(\frac{2}{3}\right)$
8. Let $g(x)$ be a cubic polynomial having local maximum at $x = -1$ and $g'(x)$ has a local minimum at $x = 1$. If $g(-1) = 10, g(3) = -22$, then :
- perpendicular distance between its two horizontal tangents is 12
 - perpendicular distance between its two horizontal tangents is 32
 - $g(x) = 0$ has atleast one real root lying in interval $(-1, 0)$
 - $g(x) = 0$, has 3 distinct real roots
9. The function $f(x) = 2x^3 - 3(\lambda + 2)x^2 + 2\lambda x + 5$ has a maximum and a minimum for :
- $\lambda \in (-4, \infty)$
 - $\lambda \in (-\infty, 0)$
 - $\lambda \in (-3, 3)$
 - $\lambda \in (1, \infty)$
10. The function $f(x) = 1 + x \ln\left(x + \sqrt{1+x^2}\right) - \sqrt{1-x^2}$ is :
- strictly increasing $\forall x \in (0, 1)$
 - strictly decreasing $\forall x \in (-1, 0)$
 - strictly decreasing for $x \in (-1, 0)$
 - strictly decreasing for $x \in (0, 1)$
11. Let m and n be positive integers and $x, y > 0$ and $x + y = k$, where k is constant. Let $f(x, y) = x^m y^n$, then :
- $f(x, y)$ is maximum when $x = \frac{mk}{m+n}$
 - $f(x, y)$ is maximum where $x = y$
 - maximum value of $f(x, y)$ is $\frac{m^n n^m k^{m+n}}{(m+n)^{m+n}}$
 - maximum value of $f(x, y)$ is $\frac{k^{m+n} m^m n^n}{(m+n)^{m+n}}$
12. The straight line which is both tangent and normal to the curve $x = 3t^2, y = 2t^3$ is :
- $y + \sqrt{3}(x-1) = 0$
 - $y - \sqrt{3}(x-1) = 0$
 - $y + \sqrt{2}(x-2) = 0$
 - $y - \sqrt{2}(x-2) = 0$
13. A curve is such that the ratio of the subnormal at any point to the sum of its co-ordinates is equal to the ratio of the ordinate of this point to its abscissa. If the curve passes through $(1, 0)$, then possible equation of the curve(s) is :
- $y = x \ln x$
 - $y = \frac{\ln x}{x}$
 - $y = \frac{2(x-1)}{x^2}$
 - $y = \frac{1-x^2}{2x}$

14. A parabola of the form $y = ax^2 + bx + c$ ($a > 0$) intersects the graph of $f(x) = \frac{1}{x^2 - 4}$. The number of possible distinct intersection(s) of these graph can be :
- (a) 0 (b) 2 (c) 3 (d) 4
15. Gradient of the line passing through the point (2, 8) and touching the curve $y = x^3$, can be :
- (a) 3 (b) 6 (c) 9 (d) 12
16. The equation $x + \cos x = a$ has exactly one positive root, then :
- (a) $a \in (0, 1)$ (b) $a \in (2, 3)$ (c) $a \in (1, \infty)$ (d) $a \in (-\infty, 1)$
17. Given that $f(x)$ is a non-constant linear function. Then the curves :
- (a) $y = f(x)$ and $y = f^{-1}(x)$ are orthogonal
- (b) $y = f(x)$ and $y = f^{-1}(-x)$ are orthogonal
- (c) $y = f(-x)$ and $y = f^{-1}(x)$ are orthogonal
- (d) $y = f(-x)$ and $y = f^{-1}(-x)$ are orthogonal
18. Let $f(x) = \int_0^x e^{t^3} (t^2 - 1)t^2(t + 1)^{2011}(t - 2)^{2012}$ at ($x > 0$) then :
- (a) The number of point of inflections is atleast 1
- (b) The number of point of inflections is 0
- (c) The number of point of local maxima is 1
- (d) The number of point of local minima is 1
19. Let $f(x) = \sin x + ax + b$. Then $f(x) = 0$ has :
- (a) only one real root which is positive if $a > 1, b < 0$
- (b) only one real root which is negative if $a > 1, b > 0$
- (c) only one real root which is negative if $a < -1, b < 0$
- (d) only one real root which is positive if $a < -1, b < 0$
20. Which of the following graphs represent function whose derivatives have a maximum in the interval (0, 1) ?





21. Consider $f(x) = \sin^5 x + \cos^5 x - 1$, $x \in \left[0, \frac{\pi}{2}\right]$, which of the following is/are correct ?

- (a) f is strictly decreasing in $\left[0, \frac{\pi}{4}\right]$
- (b) f is strictly increasing in $\left[\frac{\pi}{4}, \frac{\pi}{2}\right]$
- (c) There exist a number 'c' in $\left(0, \frac{\pi}{2}\right)$ such that $f'(c) = 0$
- (d) The equation $f(x) = 0$ has only two roots in $\left[0, \frac{\pi}{2}\right]$

22. Let $f(x) = \begin{cases} x^{2\alpha+1} \ln x & ; x > 0 \\ 0 & ; x = 0 \end{cases}$

If $f(x)$ satisfies Rolle's theorem in interval $[0, 1]$, then α can be :

- (a) $-\frac{1}{2}$
- (b) $-\frac{1}{3}$
- (c) $-\frac{1}{4}$
- (d) -1

23. Which of the following is/are true for the function $f(x) = \int_0^x \frac{\cos t}{t} dt$ ($x > 0$) ?

- (a) $f(x)$ is monotonically increasing in $\left((4n-1)\frac{\pi}{2}, (4n+1)\frac{\pi}{2}\right) \forall n \in \mathbb{N}$
- (b) $f(x)$ has a local minima at $x = (4n-1)\frac{\pi}{2} \forall n \in \mathbb{N}$
- (c) The points of inflection of the curve $y = f(x)$ lie on the curve $x \tan x + 1 = 0$
- (d) Number of critical points of $y = f(x)$ in $(0, 10\pi)$ are 19

24. Let $F(x) = (f(x))^2 + (f'(x))^2$, $F(0) = 6$, where $f(x)$ is a thrice differentiable function such that $|f(x)| \leq 1 \forall x \in [-1, 1]$, then choose the correct statement(s)

- (a) there is atleast one point in each of the intervals $(-1, 0)$ and $(0, 1)$ where $|f'(x)| \leq 2$
- (b) there is atleast one point in each of the intervals $(-1, 0)$ and $(0, 1)$ where $F(x) \leq 5$
- (c) there is no point of local maxima of $F(x)$ in $(-1, 1)$
- (d) for some $c \in (-1, 1)$, $F(c) \geq 6$, $F'(c) = 0$ and $F''(c) \leq 0$

25. Let $f(x) = \begin{cases} x^3 + x^2 - 10x; & -1 \leq x < 0 \\ \sin x; & 0 \leq x < \frac{\pi}{2} \\ 1 + \cos x; & \frac{\pi}{2} \leq x \leq \pi \end{cases}$

then $f(x)$ has :

- (a) local maximum at $x = \frac{\pi}{2}$ (b) local minimum at $x = \frac{\pi}{2}$
 (c) absolute maximum at $x = 0$ (d) absolute maximum at $x = -1$

26. Minimum distance between the curves $y^2 = x - 1$ and $x^2 = y - 1$ is equal to :

- (a) $\frac{\sqrt{2}}{4}$ (b) $\frac{3\sqrt{2}}{4}$ (c) $\frac{5\sqrt{2}}{4}$ (d) $\frac{7\sqrt{2}}{4}$

27. For the equation $\frac{e^{-x}}{1+x} = \lambda$ which of the following statement(s) is/are correct ?

- (a) When $\lambda \in (0, \infty)$ equation has 2 real and distinct roots
 (b) When $\lambda \in (-\infty, -e^2)$ equation has 2 real and distinct roots
 (c) When $\lambda \in (0, \infty)$ equation has 1 real root
 (d) When $\lambda \in (-e, 0)$ equation has no real root

28. If $y = mx + 5$ is a tangent to the curve $x^3y^3 = ax^3 + by^3$ at $P(1, 2)$, then

- (a) $a + b = \frac{18}{5}$ (b) $a > b$ (c) $a < b$ (d) $a + b = \frac{19}{5}$

29. If $(f(x) - 1)(x^2 + x + 1)^2 - (f(x) + 1)(x^4 + x^2 + 1) = 0$

$\forall x \in \mathbb{R} - \{0\}$ and $f(x) \neq \pm 1$, then which of the following statement(s) is/are correct ?

- (a) $|f(x)| \geq 2 \forall x \in \mathbb{R} - \{0\}$ (b) $f(x)$ has a local maximum at $x = -1$
 (c) $f(x)$ has a local minimum at $x = 1$ (d) $\int_{-\pi}^{\pi} (\cos x) f(x) dx = 0$

Answers

1.	(c, d)	2.	(a, c, d)	3.	(a, c)	4.	(a, c)	5.	(a, b, c, d)	6.	(b, c, d)
7.	(c, d)	8.	(b, d)	9.	(a, b, c, d)	10.	(a, c)	11.	(a, d)	12.	(c, d)
13.	(a, d)	14.	(b, c, d)	15.	(a, d)	16.	(b, c)	17.	(b, c)	18.	(a, d)
19.	(a, b, c)	20.	(a, b)	21.	(a, b, c, d)	22.	(b, c)	23.	(a, b, c)	24.	(a, b, d)
25.	(a, d)	26.	(b)	27.	(b, c, d)	28.	(a, d)	29.	(a, b, c, d)		

Exercise-3 : Comprehension Type Problems

Paragraph for Question Nos. 1 to 2

Let $y = f(x)$ such that $xy = x + y + 1$, $x \in \mathbb{R} - \{-1\}$ and $g(x) = xf(x)$

- The minimum value of $g(x)$ is :
 (a) $3 - \sqrt{2}$ (b) $3 + \sqrt{2}$ (c) $3 - 2\sqrt{2}$ (d) $3 + 2\sqrt{2}$
- There exists two values of x, x_1 and x_2 where $g'(x) = \frac{1}{2}$, then $|x_1| + |x_2| =$
 (a) 1 (b) 2 (c) 4 (d) 5

Paragraph for Question Nos. 3 to 5

Let $f(x) = \begin{cases} 1-x & ; 0 \leq x \leq 1 \\ 0 & ; 1 < x \leq 2 \\ (2-x)^2 & ; 2 < x \leq 3 \end{cases}$ and $g(x) = \int_0^x f(t) dt$.

Let the tangent to the curve $y = g(x)$ at point P whose abscissa is $\frac{5}{2}$ cuts x -axis in point Q .

Let the perpendicular from point Q on x -axis meets the curve $y = g(x)$ in point R .

- $g(1) =$
 (a) 0 (b) $\frac{1}{2}$ (c) 1 (d) 2
- Equation of tangent to the curve $y = g(x)$ at P is :
 (a) $3y = 12x + 1$ (b) $3y = 12x - 1$ (c) $12y = 3x - 1$ (d) $12y = 3x + 1$
- If θ be the angle between tangents to the curve $y = g(x)$ at point P and R ; then $\tan \theta$ equals to :
 (a) $\frac{5}{6}$ (b) $\frac{5}{14}$ (c) $\frac{5}{7}$ (d) $\frac{5}{12}$

Paragraph for Question Nos. 6 to 8

Let $f(x) < 0 \forall x \in (-\infty, 0)$ and $f(x) > 0 \forall x \in (0, \infty)$ also $f(0) = 0$. Again $f'(x) < 0 \forall x \in (-\infty, -1)$ and $f'(x) > 0 \forall x \in (-1, \infty)$ also $f'(-1) = 0$ given $\lim_{x \rightarrow -\infty} f(x) = 0$ and $\lim_{x \rightarrow \infty} f(x) = \infty$ and function is twice differentiable.

- If $f''(x) > 0 \forall x \in (-1, \infty)$ and $f'(0) = 1$ then number of solutions of equation $f(x) = x$ is :
 (a) 2 (b) 3 (c) 4 (d) None of these
- If $f''(x) < 0 \forall x \in (0, \infty)$ and $f'(0) = 1$ then number of solutions of equation $f(x) = x^2$ is :
 (a) 1 (b) 2 (c) 3 (d) 4

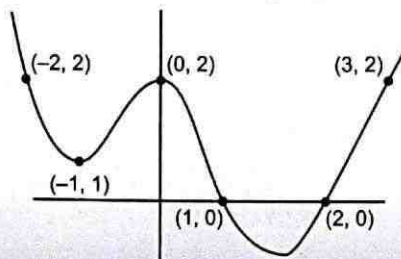
8. The minimum number of points where $f''(x)$ is zero is :

(a) 1 (b) 2 (c) 3 (d) 4

Paragraph for Question Nos. 9 to 11

In the given figure graph of :

$$y = p(x) = x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_n \text{ is given.}$$



9. The product of all imaginary roots of $p(x) = 0$ is :
 (a) -2 (b) -1 (c) $-1/2$ (d) none of these
10. If $p(x) + k = 0$ has 4 distinct real roots $\alpha, \beta, \gamma, \delta$ then $[\alpha] + [\beta] + [\gamma] + [\delta]$, (where $[\cdot]$ denotes greatest integer function) is equal to :
 (a) -1 (b) -2 (c) 0 (d) 1
11. The minimum number of real roots of equation $(p'(x))^2 + p(x)p''(x) = 0$ are :
 (a) 3 (b) 4 (c) 5 (d) 6

Paragraph for Question Nos. 12 to 14

The differentiable function $y = f(x)$ has a property that the chord joining any two points $A(x_1, f(x_1))$ and $B(x_2, f(x_2))$ always intersects y-axis at $(0, 2x_1x_2)$. Given that $f(1) = -1$, then :

12. $\int_0^{1/2} f(x) dx$ is equal to :
 (a) $\frac{1}{6}$ (b) $\frac{1}{8}$ (c) $\frac{1}{12}$ (d) $\frac{1}{24}$
13. The largest interval in which $f(x)$ is monotonically increasing, is :
 (a) $\left(-\infty, \frac{1}{2}\right]$ (b) $\left[\frac{-1}{2}, \infty\right)$ (c) $\left(-\infty, \frac{1}{4}\right]$ (d) $\left[\frac{-1}{4}, \infty\right)$
14. In which of the following intervals, the Rolle's theorem is applicable to the function $F(x) = f(x) + x$?
 (a) $[-1, 0]$ (b) $[0, 1]$ (c) $[-1, 1]$ (d) $[0, 2]$

Paragraph for Question Nos. 15 to 16

Let $f(x) = 1 + \int_0^1 (xe^y + ye^x) f(y) dy$ where x and y are independent variables.

- 15.** If complete solution set of ' x ' for which function $h(x) = f(x) + 3x$ is strictly increasing is $(-\infty, k)$ then $\left[\frac{4}{3}e^k\right]$ equals to : (where $[\cdot]$ denotes greatest integer function):

(a) 1 (b) 2 (c) 3 (d) 4

- 16.** If acute angle of intersection of the curves $\frac{x}{2} + \frac{y}{3} + \frac{1}{3} = 0$ and $y = f(x)$ be θ then $\tan \theta$ equals to :

(a) $\frac{8}{25}$ (b) $\frac{16}{25}$ (c) $\frac{14}{25}$ (d) $\frac{4}{5}$

Answers

1. (d)	2. (c)	3. (b)	4. (c)	5. (b)	6. (d)	7. (b)	8. (a)	9. (d)	10. (a)
11. (b)	12. (d)	13. (c)	14. (b)	15. (c)	16. (a)				

**Exercise-4 : Matching Type Problems**

1. Column-I gives pair of curves and column-II gives the angle θ between the curves at their intersection point.

Column-I		Column-II	
(A)	$y = \sin x, y = \cos x$	(P)	$\frac{\pi}{4}$
(B)	$x^2 = 4y, y = \frac{8}{x^2 + 4}$	(Q)	$\frac{\pi}{2}$
(C)	$\frac{x^2}{18} + \frac{y^2}{8} = 1, x^2 - y^2 = 5$	(R)	$\tan^{-1} 3$
(D)	$xy = 1, x^2 - y^2 = 5$	(S)	$\tan^{-1} 5$
		(T)	$\tan^{-1}(2\sqrt{2})$

2.

Column-I		Column-II	
(A)	$(\sin^{-1} x)^{\cos^{-1} x} - (\cos^{-1} x)^{\sin^{-1} x} \forall x \in (\cos 1, \sin 1)$	(P)	Always positive
(B)	$(\cos x)^{\sin x} - (\sin x)^{\cos x} \forall x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$	(Q)	Always negative
(C)	$(\sin x)^{\sin x} - (\cos x)^{\sin x} \forall x \in \left(0, \frac{\pi}{2}\right)$	(R)	May be positive or negative for some values of x
(D)	$(\ln(\ln x))^{\ln(\ln x)} - (\ln x)^{\ln x} \forall x \in (e^e, \infty)$	(S)	May result in zero for some of values of x
		(T)	Indeterminate

3. Let $f(x) = \frac{x^3 - 4}{(x-1)^3} \forall x \neq 1$, $g(x) = \frac{x^4 - 2x^2}{4} \forall x \in \mathbb{R}$, $h(x) = \frac{x^3 + 4}{(x+1)^3} \forall x \neq -1$,

Column-I		Column-II	
(A)	The number of possible distinct real roots of equation $f(x) = c$ where $c \geq 4$ can be	(P)	0
(B)	The number of possible distinct real roots of equation $g(x) = c$, where $c \geq 0$ can be	(Q)	1
(C)	The number of possible distinct real roots of equation $h(x) = c$, where $c \geq 1$ can be	(R)	2

(D)	The number of possible distinct real roots of equation $g(x) = c$ where $-1 < c < 0$ can be	(S)	3
		(T)	4

4.

	Column-I		Column-II
(A)	If α, β, γ are roots of $x^3 - 3x^2 + 2x + 4 = 0$ and $y = 1 + \frac{\alpha}{x-\alpha} + \frac{\beta x}{(x-\alpha)(x-\beta)} + \frac{\gamma x^2}{(x-\alpha)(x-\beta)(x-\gamma)}$ then value of y at $x = 2$ is :	(P)	2
(B)	If $x^3 + ax + 1 = 0$ and $x^4 + ax + 1 = 0$ have a common roots then the value of $ a $ can be equal to	(Q)	3
(C)	The number of local maximas of the function $x^2 + 4\cos x + 5$ is more than	(R)	4
(D)	If $f(x) = 2 x ^3 + 3x^2 - 12 x + 1$, where $x \in [-1, 2]$ then greatest value of $f(x)$ is more than	(S)	5
		(T)	0

5.

	Column-I		Column-II
(A)	Maximum value of $f(x) = \log_2 \left(\frac{4}{\sqrt{x+2} + \sqrt{2-x}} \right)$	(P)	0
(B)	The value of $\left[4 \sum_{n=1}^{\infty} \cot^{-1} \left(1 + \sum_{k=1}^n 2k \right) \right] =$ ([.] represent greatest integer function)	(Q)	1
(C)	Let $f(x) = x \sin \pi x$, $x > 0$ then number of points in $(0, 2)$ where $f'(x)$ vanishes, is	(R)	2
(D)	$\lim_{x \rightarrow 0^+} \left[\frac{x}{e^x - 1} \right] =$ ([.] represent greatest integer function)	(S)	3

6. Consider the function $f(x) = \frac{\ln x}{8} - ax + x^2$ and $a \geq 0$ is a real constant :

Column-I		Column-II	
(A)	$f(x)$ gives a local maxima at	(P)	$a = 1; x = \frac{1}{4}$
(B)	$f(x)$ gives a local minima at	(Q)	$a > 1; x = \frac{a - \sqrt{a^2 - 1}}{4}$
(C)	$f(x)$ gives a point of inflection for	(R)	$0 \leq a < 1$
(D)	$f(x)$ is strictly increasing for all $x \in \mathbb{R}^+$	(S)	$a > 1; x = \frac{a + \sqrt{a^2 - 1}}{4}$

7. The function $f(x) = \sqrt{ax^3 + bx^2 + cx + d}$ has its non-zero local minimum and maximum values at $x = -2$ and $x = 2$ respectively. If 'a' is one of the root of $x^2 - x - 6 = 0$, then match the following :

Column-I		Column-II	
(A)	The value of 'a' is	(P)	0
(B)	The value of 'b' is	(Q)	24
(C)	The value of 'c' is	(R)	Greater than 32
(D)	The value of 'd' is	(S)	-2

8.

Column-I		Column-II	
(A)	The ratio of altitude to the radius of the cylinder of maximum volume that can be inscribed in a given sphere is	(P)	$\frac{1}{\sqrt{2}}$
(B)	The ratio of radius to the altitude of the cone of the greatest volume which can be inscribed in a given sphere is	(Q)	$\sqrt{2}$
(C)	The cone circumscribing the sphere of radius 'r' has the maximum volume if its semi vertical angle is θ , then $33 \sin \theta =$	(R)	$\frac{32}{3}$
(D)	The greatest value of $x^3 y^4$ if $2x + 3y = 7$, $x \geq 0, y \geq 0$ is	(S)	11

Answers

1. $A \rightarrow T$; $B \rightarrow R$; $C \rightarrow Q$; $D \rightarrow Q$
2. $A \rightarrow R, S$; $B \rightarrow Q$; $C \rightarrow R, S$; $D \rightarrow Q$
3. $A \rightarrow Q, R$; $B \rightarrow R, S$; $C \rightarrow Q, R, S$; $D \rightarrow P, R, T$
4. $A \rightarrow P$; $B \rightarrow P$; $C \rightarrow T$; $D \rightarrow P, Q, R, T$
5. $A \rightarrow Q$; $B \rightarrow S$; $C \rightarrow R$; $D \rightarrow P$
6. $A \rightarrow Q$; $B \rightarrow S$; $C \rightarrow P$; $D \rightarrow R$
7. $A \rightarrow S$; $B \rightarrow P$; $C \rightarrow Q$; $D \rightarrow R$
8. $A \rightarrow Q$; $B \rightarrow P$; $C \rightarrow S$; $D \rightarrow R$

Exercise-5 : Subjective Type Problems

1. A conical vessel is to be prepared out of a circular sheet of metal of unit radius. In order that the vessel has maximum volume, the sectorial area that must be removed from the sheet is A_1 and the area of the given sheet is A_2 . If $\frac{A_2}{A_1} = m + \sqrt{n}$, where $m, n \in \mathbb{N}$, then $m + n$ is equal to.
2. On $[1, e]$, the least and greatest values of $f(x) = x^2 \ln x$ are m and M respectively, then $[\sqrt{M + m}]$ is : (where $[\]$ denotes greatest integer function)
3. If $f(x) = \frac{px}{e^x} - \frac{x^2}{2} + x$ is a decreasing function for every $x \leq 0$. Find the least value of p^2 .
4. Let $f(x) = \begin{cases} xe^{ax} & , x \leq 0 \\ x + ax^2 - x^3 & , x > 0 \end{cases}$. Where a is a positive constant. The interval in which $f'(x)$ is increasing is $\left[\frac{k}{a}, \frac{a}{l}\right]$. Then $k + l$ is equal to
5. Find sum of all possible values of the real parameter 'b' if the difference between the largest and smallest values of the function $f(x) = x^2 - 2bx + 1$ in the interval $[0, 1]$ is 4.
6. Let ' θ ' be the angle in radians between the curves $\frac{x^2}{36} + \frac{y^2}{4} = 1$ and $x^2 + y^2 = 12$. If $\theta = \tan^{-1}\left(\frac{a}{\sqrt{3}}\right)$; Find the value of a .
7. Let set of all possible values of λ such that $f(x) = e^{2x} - (\lambda + 1)e^x + 2x$ is monotonically increasing for $\forall x \in \mathbb{R}$ is $(-\infty, k]$. Find the value of k .
8. Let a, b, c and d be non-negative real number such that $a^5 + b^5 \leq 1$ and $c^5 + d^5 \leq 1$. Find the maximum value of $a^2c^3 + b^2d^3$.
9. There is a point (p, q) on the graph of $f(x) = x^2$ and a point (r, s) on the graph of $g(x) = -8/x$, where $p > 0$ and $r > 0$. If the line through (p, q) and (r, s) is also tangent to both the curves at these points respectively, then find the value of $(p + r)$.
10. $f(x) = \max |2 \sin y - x|$ where $y \in \mathbb{R}$ then determine the minimum value of $f(x)$.
11. Let $f(x) = \int_0^x ((a-1)(t^2 + t + 1)^2 - (a+1)(t^4 + t^2 + 1)) dt$. Then the total number of integral values of ' a ' for which $f'(x) = 0$ has no real roots is
12. The number of real roots of the equation $x^{2013} + e^{2014x} = 0$ is
13. Let the maximum value of expression $y = \frac{x^4 - x^2}{x^6 + 2x^3 - 1}$ for $x > 1$ is $\frac{p}{q}$ where p and q are relatively prime positive integers. Find the value of $(p + q)$.

14. The least positive value of the parameter 'a' for which there exists atleast one line that is tangent to the graph of the curve $y = x^3 - ax$, at one point and normal to the graph at another point is $\frac{p}{q}$; where p and q are relatively prime positive integers. Find product pq .
15. Let $f(x) = x^2 + 2x - t^2$ and $f(x) = 0$ has two roots $\alpha(t)$ and $\beta(t)$ ($\alpha < \beta$) where t is a real parameter. Let $I(t) = \int_{\alpha}^{\beta} f(x) dx$. If the maximum value of $I(t)$ be λ and $|\lambda| = \frac{p}{q}$ where p and q are relatively prime positive integers. Find the product (pq).
16. A tank contains 100 litres of fresh water. A solution containing 1 gm/litre of salt runs into the tank at the rate of 1 lit/min. The homogenised mixture is pumped out of the tank at the rate of 3 lit/min. If T be the time when the amount of salt in the tank is maximum. Find $[T]$ (where $[\cdot]$ denotes greatest integer function)
17. If $f(x)$ is continuous and differentiable in $[-3, 9]$ and $f'(x) \in [-2, 8] \forall x \in (-3, 9)$. Let N be the number of divisors of the greatest possible value of $f(9) - f(-3)$, then find the sum of digits of N .
18. It is given that $f(x)$ is defined on R satisfying $f(1) = 1$ and for $\forall x \in R$, $f(x+5) \geq f(x) + 5$ and $f(x+1) \leq f(x) + 1$. If $g(x) = f(x) + 1 - x$, then $g(2002) =$
19. The number of normals to the curve $3y^3 = 4x$ which passes through the point $(0, 1)$ is
20. Find the number of real root(s) of the equation $ae^x = 1 + x + \frac{x^2}{2}$; where a is positive constant.
21. Let $f(x) = ax + \cos 2x + \sin x + \cos x$ is defined for $\forall x \in R$ and $a \in R$ and is strictly increasing function. If the range of a is $\left[\frac{m}{n}, \infty\right)$, then find the minimum value of $(m - n)$.
22. If p_1 and p_2 are the lengths of the perpendiculars from origin on the tangent and normal drawn to the curve $x^{2/3} + y^{2/3} = 6^{2/3}$ respectively. Find the value of $\sqrt{4p_1^2 + p_2^2}$.

Answers

1.	9	2.	2	3.	1	4.	1	5.	1	6.	2	7.	3
8.	1	9.	5	10.	2	11.	3	12.	1	13.	7	14.	12
15.	12	16.	27	17.	3	18.	1	19.	1	20.	1	21.	9
22.	6												



INDEFINITE AND DEFINITE INTEGRATION

Exercise-1 : Single Choice Problems

1. $\int a^x \left(\ln x + \ln a \cdot \ln \left(\frac{x}{e} \right)^x \right) dx =$

(a) $a^x \ln \left(\frac{e}{x} \right)^{2x} + C$

☒ (b) $a^x \ln \left(\frac{x}{e} \right)^x + C$

(c) $a^x + \ln \left(\frac{x}{e} \right)^x + C$

(d) None of these

2. The value of :

$$\lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{n}\sqrt{n+1}} + \frac{1}{\sqrt{n}\sqrt{n+2}} + \frac{1}{\sqrt{n}\sqrt{n+3}} + \dots + \frac{1}{\sqrt{n}\sqrt{2n}} \right) \text{ is :}$$

(a) $\sqrt{2} - 1$

☒ (b) $2(\sqrt{2} - 1)$

(c) $\sqrt{2} + 1$

(d) $2(\sqrt{2} + 1)$

3. If $\int \frac{\sin x}{\sin(x-\alpha)} dx = Ax + B \log \sin(x-\alpha) + C$, then value of (A, B) is :

(a) $(\sin \alpha, \cos \alpha)$

☒ (b) $(\cos \alpha, \sin \alpha)$

(c) $(-\sin \alpha, \cos \alpha)$

(d) $(-\cos \alpha, \sin \alpha)$

4. The value of the integral $\int_0^2 \frac{\log(x^2 + 2)}{(x+2)^2} dx$ is :

(a) $\frac{\sqrt{2}}{3} \tan^{-1} \sqrt{2} + \frac{5}{12} \log 2 - \frac{1}{4} \log 3$

(b) $\frac{\sqrt{2}}{3} \tan^{-1} \sqrt{2} - \frac{5}{12} \log 2 - \frac{1}{12} \log 3$

(c) $\frac{\sqrt{2}}{3} \tan^{-1} \sqrt{2} + \frac{5}{12} \log 2 + \frac{1}{12} \log 3$

☒ (d) $\frac{\sqrt{2}}{3} \tan^{-1} \sqrt{2} - \frac{5}{12} \log 2 + \frac{1}{12} \log 3$

5. If $I_1 = \int_0^1 \frac{1+x^8}{1+x^4} dx$ and $I_2 = \int_0^1 \frac{1+x^9}{1+x^3} dx$, then :

(a) $I_1 > 1, I_2 < 1$

(b) $I_1 < 1, I_2 > 1$

(c) $1 < I_1 < I_2$

☒ (d) $I_2 < I_1 < 1$

6. Let $f: (0, 1) \rightarrow (0, 1)$ be a differentiable function such that $f'(x) \neq 0$ for all $x \in (0, 1)$ and

$$f\left(\frac{1}{2}\right) = \frac{\sqrt{3}}{2}. \text{ Suppose for all } x, \lim_{t \rightarrow x} \left(\frac{\int_0^t \sqrt{1 - (f(s))^2} ds - \int_0^x \sqrt{1 - (f(s))^2} ds}{f(t) - f(x)} \right) = f(x). \text{ Then the value}$$

of $f\left(\frac{1}{4}\right)$ belongs to :

- (a) $\left\{\frac{\sqrt{7}}{4}, \frac{\sqrt{15}}{4}\right\}$ (b) $\left\{\frac{\sqrt{7}}{3}, \frac{\sqrt{15}}{3}\right\}$ (c) $\left\{\frac{\sqrt{7}}{2}, \frac{\sqrt{15}}{2}\right\}$ (d) $\{\sqrt{7}, \sqrt{15}\}$

7. If $f(\theta) = \frac{4}{3}(1 - \cos^6 \theta - \sin^6 \theta)$, then

$$\lim_{n \rightarrow \infty} \frac{1}{n} \left[\sqrt{f\left(\frac{1}{n}\right)} + \sqrt{f\left(\frac{2}{n}\right)} + \sqrt{f\left(\frac{3}{n}\right)} + \dots + \sqrt{f\left(\frac{n}{n}\right)} \right] =$$

- (a) $\frac{1 - \cos 1}{2}$ (b) $1 - \cos 2$ (c) $\frac{\sin 2}{2}$ (d) $\frac{1 - \cos 2}{2}$

8. The value of $\int \frac{(x^6 - x^3)}{(2x^3 + 1)^3} dx$ is equal to :

- (a) $-\frac{1}{6}$ (b) $-\frac{1}{12}$ (c) $-\frac{1}{18}$ (d) $-\frac{1}{36}$

9. $2 \int_0^{\frac{\sqrt{2}}{2}} \frac{\sin^{-1} x}{x} dx - \int_0^1 \frac{\tan^{-1} x}{x} dx =$

- (a) $\frac{\pi}{8} \ln 2$ (b) $\frac{\pi}{4} \ln 2$ (c) $\frac{\pi}{2\sqrt{2}} \ln 2$ (d) $\frac{\pi}{2} \ln 2$

10. Let $f(x)$ be a differentiable function such that $f(x) = x^2 + \int_0^x e^{-t} f(x-t) dt$, then $\int_0^1 f(x) dx =$

- (a) $\frac{1}{3}$ (b) $\frac{1}{4}$ (c) $\frac{7}{12}$ (d) $\frac{5}{12}$

11. If $f'(x) = f(x) + \int_0^1 f(x) dx$ and given $f(0) = 1$, then $\int f(x) dx$ is equal to :

- (a) $\frac{2}{3-e} e^x + \left(\frac{3-e}{1-e}\right)x + C$ (b) $\frac{2}{3-e} e^x + \left(\frac{1-e}{3-e}\right)x + C$
 (c) $\frac{3}{2-e} e^x + \left(\frac{1+e}{3+e}\right)x + C$ (d) $\frac{2}{2-e} e^x + \left(\frac{1-e}{3+e}\right)x + C$

(where C is an arbitrary constant.)

12. For any $x \in \mathbb{R}$, and f be a continuous function. Let $I_1 = \int_{\sin^2 x}^{1+\cos^2 x} t f(t(2-t)) dt$, $I_2 = \int_{\sin^2 x}^{1+\cos^2 x} f(t(2-t)) dt$,

then $I_1 =$

- (a) I_2 (b) $\frac{1}{2}I_2$ (c) $2I_2$ (d) $3I_2$

13. If the integral $\int \frac{5 \tan x dx}{\tan x - 2} = x + a \ln |\sin x - 2 \cos x| + C$, then 'a' is equal to :

- (a) 1 (b) 2 (c) -1 (d) -2

14. $\int \frac{(2+\sqrt{x}) dx}{(x+1+\sqrt{x})^2}$ is equal to :

- (a) $\frac{x}{x+\sqrt{x}+1} + C$ (b) $\frac{2x}{x+\sqrt{x}+1} + C$
 (c) $\frac{-2x}{x+\sqrt{x}+1} + C$ (d) $\frac{-x}{x+\sqrt{x}+1} + C$

(where C is an arbitrary constant.)

15. Evaluate $\int \frac{\left(\sqrt[3]{x+\sqrt{2-x^2}}\right)\left(\sqrt[6]{1-x\sqrt{2-x^2}}\right) dx}{\sqrt[3]{1-x^2}}$; $x \in (0, 1)$:

- (a) $2^{\frac{1}{6}}x + C$ (b) $2^{\frac{1}{12}}x + C$
 (c) $2^{\frac{1}{3}}x + C$ (d) None of these

16. $\int \frac{dx}{\sqrt{1-\tan^2 x}} = \frac{1}{\lambda} \sin^{-1}(\lambda \sin x) + C$, then $\lambda =$

- (a) $\sqrt{2}$ (b) $\sqrt{3}$ (c) 2 (d) $\sqrt{5}$

17. $\int \frac{dx}{\sqrt[3]{x^{5/2}(x+1)^{7/2}}}$ is equal to :

- (a) $-\left(\frac{x+1}{x}\right)^{1/6} + C$ (b) $6\left(\frac{x+1}{x}\right)^{-1/6} + C$
 (c) $\left(\frac{x}{x+1}\right)^{5/6} + C$ (d) $-\left(\frac{x}{x+1}\right)^{5/6} + C$

18. If $I_n = \int (\sin x)^n dx$; $n \in \mathbb{N}$, then $5I_4 - 6I_6$ is equal to :

- (a) $\sin x \cdot (\cos x)^5 + C$ (b) $\sin 2x \cos 2x + C$
 (c) $\frac{\sin 2x}{8} [1 + \cos^2 2x - 2 \cos 2x] + C$ (d) $\frac{\sin 2x}{8} [1 + \cos^2 2x + 2 \cos 2x] + C$

19. $\int \frac{x^2}{(a+bx)^2} dx$ equals to :

- (a) $\frac{1}{b^3} \left(a+bx - a \ln|a+bx| - \frac{a^2}{a+bx} \right) + C$ (b) $\frac{1}{b^3} \left(a+bx - 2a \ln|a+bx| - \frac{a^2}{a+bx} \right) + C$
 (c) $\frac{1}{b^3} \left(a+bx + 2a \ln|a+bx| - \frac{a^2}{a+bx} \right) + C$ (d) $\frac{1}{b^3} \left(a+bx - 2a \ln|a+bx| - \frac{a^2}{a+bx} \right) + C$

20. $\int \frac{8x^{43} + 13x^{38}}{(x^{13} + x^5 + 1)^4} dx =$

- (a) $\frac{x^{39}}{3(x^{13} + x^5 + 1)^3} + C$ (b) $\frac{x^{39}}{(x^{13} + x^5 + 1)^3} + C$
 (c) $\frac{x^{39}}{5(x^{13} + x^5 + 1)^5} + C$ (d) None of these

21. $\int \left(\frac{\cos 6x + 6 \cos 4x + 15 \cos 2x + 10}{10 \cos^2 x + 5 \cos x \cos 3x + \cos x \cos 5x} \right) dx = f(x) + C$, then $f(10)$ is equal to :

- (a) 20 (b) 10 (c) $2 \sin 10$ (d) $2 \cos 10$

22. $\int (1+x-x^{-1}) e^{x+x^{-1}} dx =$

- (a) $(x+1)e^{x+x^{-1}} + C$ (b) $(x-1)e^{x+x^{-1}} + C$
 (c) $-xe^{x+x^{-1}} + C$ (d) $xe^{x+x^{-1}} + C$

23. If $\int e^x \left(\frac{2 \tan x}{1 + \tan x} + \operatorname{cosec}^2 \left(x + \frac{\pi}{4} \right) \right) dx = e^x \cdot g(x) + K$, then $g\left(\frac{5\pi}{4}\right) =$

- (a) 0 (b) 1 (c) -1 (d) 2

24. $\int e^{x \sin x + \cos x} \left(\frac{x^4 \cos^3 x - x \sin x + \cos x}{x^2 \cos^2 x} \right) dx =$

- (a) $e^{x \sin x + \cos x} \left(x - \frac{1}{\cos x} \right) + C$ (b) $e^{x \sin x + \cos x} \left(x - \frac{1}{x \cos x} \right) + C$
 (c) $e^{x \sin x + \cos x} \left(1 - \frac{1}{x \cos x} \right) + C$ (d) $e^{x \sin x + \cos x} \left(1 - \frac{x}{\cos x} \right) + C$

25. The value of the definite integral $\int_0^1 \frac{1+x+\sqrt{x+x^2}}{\sqrt{x}+\sqrt{1+x}} dx$ is :

- (a) $\frac{1}{3}(2^{1/2} - 1)$ (b) $\frac{2}{3}(2^{1/2} - 1)$
 (c) $\frac{2}{3}(2^{3/2} - 1)$ (d) $\frac{1}{3}(2^{3/2} - 1)$

26. $\int x^{x^2+1} (2 \ln x + 1) dx$
 (a) $x^{2x} + C$ (b) $x^2 \ln x + C$ (c) $x^{(x^x)} + C$ (d) $(x^x)^x + C$
27. If $\int \frac{\operatorname{cosec}^2 x - 2010}{\cos^{2010} x} dx = -\frac{f(x)}{(g(x))^{2010}} + C$; where $f\left(\frac{\pi}{4}\right) = 1$; then the number of solutions of the equation $\frac{f(x)}{g(x)} = \{x\}$ in $[0, 2\pi]$ is/are : (where $\{ \}$ represents fractional part function)
 (a) 0 (b) 1 (c) 2 (d) 3
28. $\int x^x \left((\ln x)^2 + \ln x + \frac{1}{x} \right) dx$ is equal to :
 (a) $x^x \left((\ln x)^2 - \frac{1}{x} \right) + C$ (b) $x^x (\ln x - x) + C$
 (c) $x^x \frac{(\ln x)^2}{2} + C$ (d) $x^x \ln x + C$
29. If $I = \int \frac{x^2 - 1}{x^3 \sqrt{2x^4 - 2x^2 + 1}}$ is equal to :
 (a) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{x^2} + C$ (b) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{x} + C$
 (c) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{x} + C$ (d) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{2x^2} + C$
30. $I = \int \left(\frac{\ln x - 1}{(\ln x)^2 + 1} \right)^2 dx$ is equal to :
 (a) $\frac{x}{x^2 + 1} + C$ (b) $\frac{\ln x}{(\ln x)^2 + 1} + C$ (c) $\frac{x}{1 + (\ln x)^2} + C$ (d) $e^x \left(\frac{x}{x^2 + 1} \right) + C$
31. $I = \int \frac{dx}{\sqrt[4]{(x-1)^3(x+2)^5}} = k \sqrt[4]{\frac{x-1}{x+2}} + C$, then 'k' is equal to :
 (a) $\frac{1}{3}$ (b) $\frac{2}{3}$ (c) $\frac{3}{4}$ (d) $\frac{4}{3}$
32. $\int \frac{1-x^7}{x(1+x^7)} dx = P \log|x| + Q \log|x^7 + 1| + C$, then :
 (a) $2P - 7Q = 0$ (b) $2P + 7Q = 0$ (c) $7P + 2Q = 0$ (d) $7P - 2Q = 1$
33. $I = \int \frac{\sin^8 x - \cos^8 x}{1 - 2 \sin^2 x \cos^2 x} dx$ is equal to :
 (a) $\sin 2x + C$ (b) $\frac{\sin 2x}{2} + C$ (c) $\frac{-\sin 2x}{2} + C$ (d) $-2 \sin 2x + C$

$$34. I = \int \frac{(\sin 2x)^{1/3} d(\tan^{1/3} x)}{\sin^{2/3} x + \cos^{2/3} x} =$$

$$(a) \frac{1}{2^{2/3}} \ln(1 + \tan^{1/3} x) + C$$

$$(b) \ln(1 + \tan^{2/3} x) + C$$

$$(c) 2^{1/3} \ln(1 + \tan^{2/3} x) + C$$

$$(d) \frac{1}{2^{2/3}} \ln(1 + \tan^{2/3} x) + C$$

$$35. \int \sqrt{\frac{(2012)^{2x}}{1 - (2012)^{2x}}} (2012)^{\sin^{-1}(2012)^x} dx =$$

$$(a) (\log_{2012} e)^2 (2012)^{\sin^{-1}(2012)^x} + C$$

$$(b) (\log_{2012} e)^2 (2012)^{x + \sin^{-1}(2012)^x} + C$$

$$(c) (\log_{2012} e)^2 (2012)^{\sin^{-1}(2012)^x} + C$$

$$(d) \frac{(2012)^{\sin^{-1}(2012)^x}}{(\log_{2012} e)^2} + C$$

(where C denotes arbitrary constant.)

$$36. \int \frac{(x+2) dx}{(x^2 + 3x + 3)\sqrt{x+1}} \text{ is equal to :}$$

$$(a) \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{x}{\sqrt{3(x+1)}} \right) + C$$

$$(b) \frac{2}{\sqrt{3}} \tan^{-1} \left(\sqrt{\frac{x}{3(x+1)}} \right) + C$$

$$(c) \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{\sqrt{x}}{3(x+1)} \right) + C$$

$$(d) \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{x}{\sqrt{3(x+1)}} \right) + C$$

(where C is arbitrary constant.)

$$37. \int \left(\frac{f(x)g'(x) - f'(x)g(x)}{f(x)g(x)} \right) (\log(g(x)) - \log(f(x))) dx \text{ is equal to :}$$

$$(a) \log \left(\frac{g(x)}{f(x)} \right) + C$$

$$(b) \frac{1}{2} \left(\frac{g(x)}{f(x)} \right)^2 + C$$

$$(c) \frac{1}{2} \left(\log \left(\frac{g(x)}{f(x)} \right) \right)^2 + C$$

$$(d) \log \left(\left(\frac{g(x)}{f(x)} \right)^2 \right) + C$$

$$38. \int \left(\int e^x \left(\ln x + \frac{2}{x} - \frac{1}{x^2} \right) dx \right) dx =$$

$$(a) e^x \ln x + C_1 x + C_2$$

$$(b) e^x \ln x + \frac{1}{x} + C_1 x + C_2$$

$$(c) \frac{\ln x}{x} + C_1 x + C_2$$

$$(d) \text{None of these}$$

39. Maximum value of the function $f(x) = \pi^2 \int_0^1 t \sin(x + \pi t) dt$ over all real number x :
- (a) $\sqrt{\pi^2 + 1}$ (b) $\sqrt{\pi^2 + 2}$ (c) $\sqrt{\pi^2 + 3}$ (d) $\sqrt{\pi^2 + 4}$
40. Let ' f ' is a function, continuous on $[0, 1]$ such that $f(x) \leq \sqrt{5} \forall x \in [0, 1]$ and $f(x) \leq \frac{2}{x} \forall x \in [\frac{1}{2}, 1]$ then the smallest ' a ' for which $\int_0^1 f(x) dx \leq a$ holds for all ' f ' is :
- (a) $\sqrt{5}$ (b) $\frac{\sqrt{5}}{2} + 2 \ln 2$ (c) $2 + \ln\left(\frac{\sqrt{5}}{2}\right)$ (d) $2 + 2 \ln\left(\frac{\sqrt{5}}{2}\right)$
41. Let $I_n = \int_1^{e^2} (\ln x)^n d(x^2)$, then the value of $2I_n + nI_{n-1}$ equals to :
- (a) 0 (b) $2e^2$ (c) e^2 (d) 1
42. Let a function $f: R \rightarrow R$ be defined as $f(x) = x + \sin x$. The value of $\int_0^{2\pi} f^{-1}(x) dx$ will be :
- (a) $2\pi^2$ (b) $2\pi^2 - 2$ (c) $2\pi^2 + 2$ (d) π^2
43. The value of the definite integral $\int_{-1}^1 e^{-x^4} \left(2 + \ln(x + \sqrt{x^2 + 1}) + 5x^3 - 8x^4 \right) dx$ is equal to :
- (a) $4e$ (b) $\frac{4}{e}$ (c) $2e$ (d) $\frac{2}{e}$
44. $\int_{-10}^0 \frac{\left| \frac{2[x]}{3x - [x]} \right|}{\frac{2[x]}{3x - [x]}} dx$ is equal to (where $[*]$ denotes greatest integer function.)
- (a) $\frac{28}{3}$ (b) $\frac{1}{3}$ (c) 0 (d) None of these
45. If $f(x) = \frac{x}{1 + (\ln x)(\ln x) \dots \infty} \forall x \in [1, \infty)$ then $\int_1^{2e} f(x) dx$ equals is :
- (a) $\frac{e^2 - 1}{2}$ (b) $\frac{e^2 + 1}{2}$ (c) $\frac{e^2 - 2e}{2}$ (d) None of these
46. $\int_0^4 \frac{(y^2 - 4y + 5) \sin(y - 2)}{(2y^2 - 8y + 11)} dy$ is equal to :
- (a) 0 (b) 2 (c) -2 (d) None of these

47. Let $\frac{d}{dx}F(x) = \left(\frac{e^{\sin x}}{x}\right)$, $x > 0$. If $\int_1^4 \frac{3}{x} e^{\sin x^3} dx = F(k) - F(1)$, then one of the possible values of k ,

is:

- (a) 15 (b) 16 (c) 63 (d) 64

48. Value of $\lim_{h \rightarrow 0} \frac{\int_0^{\pi+he^{-1/h}} x^2 e^{-x^2} dx - \int_0^{\pi} x^2 e^{-x^2} dx}{he^{-1/h}}$ is equal to :

- (a) $\pi(1-\pi^2)e^{-\pi^2}$ (b) $2\pi(1-\pi^2)e^{-\pi^2}$ (c) $\pi(1-\pi)e^{-\pi}$ (d) $\pi^2 e^{-\pi^2}$

49. Let $f: \mathbb{R}^+ \rightarrow \mathbb{R}$ be a differentiable function with $f(1) = 3$ and satisfying :

$$\int_1^{xy} f(t) dt = y \int_1^x f(t) dt + x \int_1^y f(t) dt \quad \forall x, y \in \mathbb{R}^+, \text{ then } f(e) =$$

- (a) 3 (b) 4 (c) 1 (d) None of these

50. If $[\cdot]$ denotes the greatest integer function, then the integral $\int_0^{\pi/2} \frac{e^{\sin x - [\sin x]} d(\sin^2 x - [\sin^2 x])}{\sin x - [\sin x]}$ is

λ , then $[\lambda - 1]$ is equal to :

- (a) 0 (b) 1 (c) 2 (d) 3

51. Calculate the reciprocal of the limit $\lim_{x \rightarrow \infty} \int_0^x x e^{t^2 - x^2} dt$

- (a) 0 (b) 1 (c) 2 (d) 3

52. Let $L = \lim_{n \rightarrow \infty} \left(\frac{(2 \cdot 1 + n)}{1^2 + n \cdot 1 + n^2} + \frac{(2 \cdot 2 + n)}{2^2 + n \cdot 2 + n^2} + \frac{(2 \cdot 3 + n)}{3^2 + n \cdot 3 + n^2} + \dots + \frac{(2 \cdot n + n)}{n^2} \right)$ then value of e^L is :

- (a) 2 (b) 3 (c) 4 (d) $\frac{3}{2}$

53. The value of the definite integral $\int_0^2 \left(\sqrt{1+x^3} + \sqrt[3]{x^2+2x} \right) dx$ is :

- (a) 4 (b) 5 (c) 6 (d) 7

54. The value of the definite integral $\int_0^{\infty} \frac{\ln x}{x^2 + 4} dx$ is :

- (a) $\frac{\pi \ln 3}{2}$ (b) $\frac{\pi \ln 2}{3}$
 (c) $\frac{\pi \ln 2}{4}$ (d) $\frac{\pi \ln 4}{3}$

55. The value of the definite integral $\int_0^{10} ((x-5) + (x-5)^2 + (x-5)^3) dx$ is :

- (a) $\frac{125}{3}$ (b) $\frac{250}{3}$ (c) $\frac{125}{6}$ (d) $\frac{250}{4}$

56. The value of definite integral $\int_0^{\infty} \frac{dx}{(1+x^9)(1+x^2)}$ equals to :

- (a) $\frac{\pi}{16}$ (b) $\frac{\pi}{8}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{2}$

57. The value of the definite integral $\int_0^{\pi/2} \left(\frac{1 + \sin 3x}{1 + 2 \sin x} \right) dx$ equals to :

- (a) $\frac{\pi}{2}$ (b) 1 (c) $\frac{1}{2}$ (d) $\frac{\pi}{4}$

58. The value of $\lim_{x \rightarrow \infty} \frac{\int_0^x (\tan^{-1} x)^2 dx}{\sqrt{x^2 + 1}} =$

- (a) $\frac{\pi^2}{16}$ (b) $\frac{\pi^2}{4}$
(c) $\frac{\pi^2}{2}$ (d) None of these

59. If $\int_0^1 \left(\sum_{r=1}^{2013} \frac{x}{x^2 + r^2} \right) \left(\prod_{r=1}^{2013} (x^2 + r^2) \right) dx = \frac{1}{2} \left[\left(\prod_{r=1}^{2013} (1 + r^2) \right) - k^2 \right]$

then $k =$

- (a) 2013 (b) 2013! (c) 2013^2 (d) 2013^{2013}

60. $f(x) = 2x - \tan^{-1} x - \ln(x + \sqrt{1+x^2})$

- (a) strictly increases $\forall x \in R$
(b) strictly increases only in $(0, \infty)$
(c) strictly decreases $\forall x \in R$
(d) strictly decreases in $(0, \infty)$ and strictly increases in $(-\infty, 0)$

61. The value of the definite integral $\int_0^{\pi/2} \frac{dx}{\tan x + \cot x + \operatorname{cosec} x + \sec x}$ is :

- (a) $1 - \frac{\pi}{4}$ (b) $\frac{\pi}{4} + 1$ (c) $\pi + \frac{1}{4}$ (d) None of these

62. The value of the definite integral $\int_3^7 \frac{\cos x^2}{\cos x^2 + \cos(10-x)^2} dx$ is :
- (a) 2 (b) 1 (c) $\frac{1}{2}$ (d) None of these

63. The value of the integral $\int_{e^{-1}}^{e^2} \left| \frac{\ln x}{x} \right| dx$ is :
- (a) $\frac{3}{2}$ (b) $\frac{5}{2}$ (c) 3 (d) 5

64. The value of $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\int_0^{\operatorname{cosec}^2 x} \operatorname{tg}(t) dt}{x^2 - \frac{\pi^2}{16}}$ is :
- (a) $\frac{2}{\pi} g(2)$ (b) $-\frac{4}{\pi} g(2)$ (c) $-\frac{16}{\pi} g(2)$ (d) $-4g(2)$

65. The value of $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n-k}{n^2} \cos \frac{4k}{n}$ equals :
- (a) $\frac{1}{4} \sin 4 + \frac{1}{16} \cos 4 - \frac{1}{16}$ (b) $\frac{1}{4} \sin 4 - \frac{1}{16} \cos 4 + \frac{1}{16}$
- (c) $\frac{1}{16} (1 - \sin 4)$ (d) $\frac{1}{16} (1 - \cos 4)$

66. For each positive integer n , define a function f_n on $[0, 1]$ as follows :

$$f_n(x) = \begin{cases} 0 & \text{if } x = 0 \\ \sin \frac{\pi}{2n} & \text{if } 0 < x \leq \frac{1}{n} \\ \sin \frac{2\pi}{2n} & \text{if } \frac{1}{n} < x \leq \frac{2}{n} \\ \sin \frac{3\pi}{2n} & \text{if } \frac{2}{n} < x \leq \frac{3}{n} \\ \sin \frac{n\pi}{2n} & \text{if } \frac{n-1}{n} < x \leq 1 \end{cases}$$

Then the value of $\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx$ is :

- (a) π (b) $\frac{\pi}{2}$
- (c) $\frac{1}{\pi}$ (d) $\frac{2}{\pi}$

67. Let n be a positive integer, then

$$\int_0^{n+1} \min\{|x-1|, |x-2|, |x-3|, \dots, |x-n|\} dx \text{ equals}$$

- (a) $\frac{(n+1)}{4}$ (b) $\frac{(n+2)}{4}$ (c) $\frac{(n+3)}{4}$ (d) $\frac{(n+4)}{4}$

68. For positive integers $k = 1, 2, 3, \dots, n$, let S_k denotes the area of ΔAOB_k (where 'O' is origin)

such that $\angle AOB_k = \frac{k\pi}{2n}$, $OA = 1$ and $OB_k = k$. The value of the $\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=1}^n S_k$ is :

- (a) $\frac{2}{\pi^2}$ (b) $\frac{4}{\pi^2}$ (c) $\frac{8}{\pi^2}$ (d) $\frac{1}{2\pi^2}$

69. If $A = \int_0^1 \prod_{r=1}^{2014} (r-x) dx$ and $B = \int_0^1 \prod_{r=0}^{2013} (r+x) dx$, then :

- (a) $A = 2B$ (b) $2A = B$ (c) $A + B = 0$ (d) $A = B$

70. If $f(x) = \left[\frac{x}{120} + \frac{x^3}{30} \right]$ defined in $[0, 3]$, then $\int_0^1 (f(x) + 2) dx =$

(where $[\cdot]$ denotes greatest integer function)

- (a) 0 (b) 1 (c) 2 (d) 4

71. If $f(x) = \int_0^{g(x)} \frac{dt}{\sqrt{1+t^3}}$, $g(x) = \int_0^{\cos x} (1 + \sin t)^2 dt$, then the value of $f'\left(\frac{\pi}{2}\right)$ is equal to :

- (a) 1 (b) -1 (c) 0 (d) $\frac{1}{2}$

72. Let $f(x) = \frac{1}{x^2} \int_0^x (4t^2 - 2f'(t)) dt$, find $9f'(4)$

- (a) 16 (b) 4 (c) 8 (d) 32

73. Evaluate $\lim_{n \rightarrow \infty} \left(\frac{1^2}{n^3 + 1^3} + \frac{2^2}{n^3 + 2^3} + \frac{3^2}{n^3 + 3^3} + \dots + \frac{4}{9n} \right)$

- (a) $\frac{1}{3} \ln 3$ (b) $\frac{\ln 9}{3}$ (c) $\frac{\ln 4}{3}$ (d) $\frac{\ln 6}{3}$

74. The value of $\int_0^{2\pi} \cos^{-1} \left(\frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \right) dx$ is :

- (a) π^2 (b) $\frac{\pi^2}{2}$ (c) $2\pi^2$ (d) π^3

75. Given a function 'g' continuous everywhere such that $\int_0^1 g(t) dt = 2$ and $g(1) = 5$.

If $f(x) = \frac{1}{2} \int_0^x (x-t)^2 g(t) dt$, then the value of $f'''(1) - f''(1)$ is :

- (a) 0 (b) 1 (c) 2 (d) 3

76. If $\int_0^{\pi} \frac{x^3 \cos^4 x \sin^2 x}{\pi^2 - 3\pi x + 3x^2} dx = \lambda \int_0^{\pi/2} \sin^2 x dx$, then the value of λ is :

- (a) $\frac{\pi}{12}$ (b) $\frac{\pi}{8}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{3}$

77. $\int_0^{\sqrt{3}} \left(\frac{1}{2} \frac{d}{dx} \left(\tan^{-1} \frac{2x}{1-x^2} \right) \right) dx$ equals to :

- (a) $\frac{\pi}{3}$ (b) $-\frac{\pi}{6}$ (c) $\frac{\pi}{2}$ (d) None of these

78. Let $y = \{x\}^{[x]}$ then the value of $\int_0^3 y dx$ equals to :

(where $\{\cdot\}$ and $[\cdot]$ denote fractional part and greatest integer function respectively.)

- (a) 1 (b) $\frac{11}{6}$ (c) 3 (d) $\frac{5}{6}$

79. $\int_0^1 \frac{\tan^{-1} x}{x} dx =$

- (a) $\int_0^{\pi/4} \frac{\sin x}{x} dx$ (b) $\int_0^{\pi/2} \frac{x}{\sin x} dx$ (c) $\frac{1}{2} \int_0^{\pi/2} \frac{x}{\sin x} dx$ (d) $\frac{1}{2} \int_0^{\pi/4} \frac{x}{\sin x} dx$

80. The value of $\int_0^{4/\pi} \left(3x^2 \sin \frac{1}{x} - x \cos \frac{1}{x} \right) dx$ is :

- (a) $\frac{8\sqrt{2}}{\pi^3}$ (b) $\frac{24\sqrt{2}}{\pi^3}$
(c) $\frac{32\sqrt{2}}{\pi^3}$ (d) None of these

81. The number of values of x satisfying the equation :

$$\int_{-1}^x \left(8t^2 + \frac{28t}{3} + 4 \right) dt = \frac{\frac{3}{2}x + 1}{\log_{(x+1)} \sqrt{x+1}}, \text{ is :}$$

- (a) 0 (b) 1 (c) 2 (d) 3

82. $\lim_{n \rightarrow \infty} \frac{1 + 2^4 + 3^4 + \dots + n^4}{n^5} - \lim_{n \rightarrow \infty} \frac{1 + 2^3 + 3^3 + \dots + n^3}{n^5}$ is :

- (a) $\frac{1}{30}$ (b) zero (c) $\frac{1}{4}$ (d) $\frac{1}{5}$

83. The value of $\lim_{x \rightarrow 0^+} \frac{\int_0^{\cos x} (\cos^{-1} t) dt}{2x - \sin 2x}$ is equal to :

- (a) 0 (b) -1 (c) $\frac{2}{3}$ (d) $-\frac{1}{4}$

84. Consider a parabola $y = \frac{x^2}{4}$ and the point $F(0, 1)$.

Let $A_1(x_1, y_1), A_2(x_2, y_2), A_3(x_3, y_3), \dots, A_n(x_n, y_n)$ are 'n' points on the parabola such that $x_k > 0$ and $\angle OFA_k = \frac{k\pi}{2n}$ ($k = 1, 2, 3, \dots, n$). Then the value of $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n FA_k$ is equal to :

- (a) $\frac{2}{\pi}$ (b) $\frac{4}{\pi}$ (c) $\frac{8}{\pi}$ (d) None of these

85. The minimum value of $f(x) = \int_0^4 e^{|x-t|} dt$ where $x \in [0, 3]$ is :

- (a) $2e^2 - 1$ (b) $e^4 - 1$ (c) $2(e^2 - 1)$ (d) $e^2 - 1$

86. If $\int_0^{\frac{\pi}{2}} \frac{\cos x}{x} dx = \frac{\pi}{2}$, then $\int_0^{\frac{\pi}{2}} \frac{\cos^3 x}{x} dx$ is equal to :

- (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$ (c) π (d) $\frac{3\pi}{2}$

87. $\int \sqrt{1 + \sin x} \left(\cos \frac{x}{2} - \sin \frac{x}{2} \right) dx = :$

- (a) $\frac{1 + \sin x}{2} + C$ (b) $(1 + \sin x)^2 + C$ (c) $\frac{1}{\sqrt{1 + \sin x}} + C$ (d) $\sin x + C$

88. If $I_n = \int_0^{\frac{\pi}{2}} \frac{\sin(2nx)}{\sin 2x} dx$, then the value of $I_{n+\frac{1}{2}}$ is equal to ($n \in I$) :

- (a) $\frac{n\pi}{2}$ (b) π (c) $\frac{\pi}{2}$ (d) 0

89. The value of function $f(x) = 1 + x + \int_1^x (\ln^2 t + 2 \ln t) dt$ where $f'(x)$ vanishes is :

- (a) $\frac{1}{e}$ (b) 0 (c) $\frac{2}{e}$ (d) $1 + \frac{2}{e}$

90. Let f be a differentiable function on $\mathbb{R}$ and satisfies $f(x) = x^2 + \int_0^x e^{-t} f(x-t) dt$; then $\int_0^1 f(x) dx$

is equal to :

- (a) $\frac{1}{3}$ (b) $\frac{1}{4}$ (c) $\frac{7}{12}$ (d) $\frac{5}{12}$

91. The value of the definite integral $\int_{-(\pi/2)}^{\pi/2} \frac{\cos^2 x}{1+5^x} dx$ equals to :

- (a) $\frac{3\pi}{4}$ (b) π (c) $\frac{\pi}{2}$ (d) $\frac{\pi}{4}$

92. $\int \left(\frac{x^2 - x + 1}{x^2 + 1} \right) e^{\cot^{-1}(x)} dx = f(x) \cdot e^{\cot^{-1}(x)} + C$

where C is constant of integration. Then $f(x)$ is equal to :

- (a) $-x$ (b) $\sqrt{1-x}$ (c) x (d) $\sqrt{1+x}$

93. $\lim_{n \rightarrow \infty} \frac{1}{n^3} (\sqrt{n^2+1} + 2\sqrt{n^2+2^2} + \dots + n\sqrt{n^2+n^2}) = :$

- (a) $\frac{3\sqrt{2}-1}{2}$ (b) $\frac{2\sqrt{2}-1}{3}$ (c) $\frac{3\sqrt{3}-1}{3}$ (d) $\frac{4\sqrt{2}-1}{2}$

94. $\int \frac{(x^3-1)}{(x^4+1)(x+1)} dx$, is :

- (a) $\frac{1}{4} \ln(1+x^4) + \frac{1}{3} \ln(1+x^3) + c$ (b) $\frac{1}{4} \ln(1+x^4) - \frac{1}{3} \ln(1+x^3) + c$
 (c) $\frac{1}{4} \ln(1+x^4) - \ln(1+x) + c$ (d) $\frac{1}{4} \ln(1+x^4) + \ln(1+x) + c$

95. The value of Limit $\frac{\int_0^{\cos x} (\cos^{-1} t) dt}{2x - \sin 2x}$ is equal to :

- (a) 0 (b) -1 (c) $\frac{2}{3}$ (d) $\frac{-1}{4}$

96. Let $f(x) = \lim_{n \rightarrow \infty} \frac{\cos x}{1 + (\tan^{-1} x)^n}$, then $\int_0^{\infty} f(x) dx =$

- (a) $\tan(\sin 1)$ (b) $\sin(\tan 1)$ (c) 0 (d) $\sin\left(\frac{\tan 1}{2}\right)$

97. The value of $\lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\frac{k}{n^2 + n + 2k} \right) =$

(a) $\frac{1}{4}$

(b) $\frac{1}{3}$

(c) $\frac{1}{2}$

(d) 1

98. The value of $\lim_{y \rightarrow 1^+} \frac{\int_1^y |t-1| dt}{\tan(y-1)}$ is :

(a) 0

(b) 1

(c) 2

(d) does not exist

99. Given that $\int \frac{dx}{(1+x^2)^n} = \frac{x}{2(n-1)(1+x^2)^{n-1}} + \frac{(2n-3)}{2(n-1)} \int \frac{dx}{(1+x^2)^{n-1}}$. Find the value of $\int_0^1 \frac{dx}{(1+x^2)^4}$: (you may or may not use reduction formula given)

(a) $\frac{11}{48} + \frac{5\pi}{64}$

(b) $\frac{11}{48} + \frac{5\pi}{32}$

(c) $\frac{1}{24} + \frac{5\pi}{64}$

(d) $\frac{1}{96} + \frac{5\pi}{32}$

100. Find the value of $\int_0^{\pi/4} (\sin x)^4 dx$:

(a) $\frac{3\pi}{16}$

(b) $\frac{3\pi}{32} - \frac{1}{4}$

(c) $\frac{3\pi}{32} - \frac{3}{4}$

(d) $\frac{3\pi}{16} - \frac{7}{8}$

101. $\int \frac{\cos 9x + \cos 6x}{2 \cos 5x - 1} dx = A \sin 4x + B \sin C$, then $A + B$ is equal to :

(Where C is constant of integration)

(a) $\frac{1}{2}$

(b) $\frac{3}{4}$

(c) 2

(d) $\frac{5}{4}$

102. $\int \frac{dx}{x^{2014} + x} = \frac{1}{p} \ln \left(\frac{x^q}{1+x^r} \right) + C$ where $p, q, r \in N$ then the value of $(p+q+r)$ equals

(Where C is constant of integration)

(a) 6039

(b) 6048

(c) 6047

(d) 6021

103. If $\int_0^1 e^{-x^2} dx = a$, then $\int_0^1 x^2 e^{-x^2} dx$ is equal to

(a) $\frac{1}{2e}(ea-1)$

(b) $\frac{1}{2e}(ea+1)$

(c) $\frac{1}{e}(ea-1)$

(d) $\frac{1}{e}(ea+1)$

104. If $f(x)$ is a continuous function for all real values of x and satisfies $\int_n^{n+1} f(x) dx = \frac{n^2}{2} \forall n \in I$, then

$\int_{-3}^5 f(|x|) dx$ is equal to :

(a) $\frac{19}{2}$

(b) $\frac{35}{2}$

(c) $\frac{17}{2}$

(d) $\frac{37}{2}$

105. If $\int \frac{dx}{x^4(1+x^3)^2} = a \ln \left| \frac{1+x^3}{x^3} \right| + \frac{b}{x^3} + \frac{c}{1+x^3} + d$, then

(where d is arbitrary constant)

(a) $a = \frac{1}{3}, b = \frac{1}{3}, c = \frac{1}{3}$

(b) $a = \frac{2}{3}, b = -\frac{1}{3}, c = \frac{1}{3}$

(c) $a = \frac{2}{3}, b = -\frac{1}{3}, c = -\frac{1}{3}$

(d) $a = \frac{2}{3}, b = \frac{1}{3}, c = -\frac{1}{3}$

106. $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}\sqrt{n+1}} + \frac{1}{\sqrt{n}\sqrt{n+2}} + \dots + \frac{1}{\sqrt{n}\sqrt{4n}}$ is equal to :

(a) 2

(b) 4

(c) $2(\sqrt{2}-1)$

(d) $2\sqrt{2}-1$

107. Let $f(x) = \int_x^2 \frac{dy}{\sqrt{1+y^3}}$. The value of the integral $\int_0^2 xf(x) dx$ is equal to :

(a) 1

(b) $\frac{1}{3}$

(c) $\frac{4}{3}$

(d) $\frac{2}{3}$

108. The value of the definite integral $\int_0^{\pi/3} \ln(1 + \sqrt{3} \tan x) dx$ equals

(a) $\frac{\pi}{3} \ln 2$

(b) $\frac{\pi}{3}$

(c) $\frac{\pi^2}{6} \ln 2$

(d) $\frac{\pi}{2} \ln 2$

109. If $\int_0^{100} f(x) dx = a$, then $\sum_{r=1}^{100} \int_0^1 (f(r-1+x) dx) =$

(a) $100a$

(b) a

(c) 0

(d) $10a$

110. The value of $\int_0^1 \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{x^{k+2} 2^k}{k!} dx$ is :

(a) $e^2 - 1$

(b) 2

(c) $\frac{e^2 - 1}{2}$

(d) $\frac{e^2 - 1}{4}$

111. Evaluate : $\int x^5 \sqrt{1+x^3} dx$.

(a) $\frac{1}{15} (1+x^3)^{5/2} - \frac{1}{9} (1+x^3)^{3/2} + c$

(b) $\frac{2}{15} (1+x^3)^{5/2} - \frac{1}{9} (1+x^3)^{3/2} + c$

(c) $\frac{2}{15} (1+x^3)^{5/2} - \frac{2}{9} (1+x^3)^{3/2} + c$

(d) $\frac{1}{15} (1+x^3)^{5/2} - \frac{2}{9} (1+x^3)^{3/2} + c$

112. If $f(x) = \int_0^x \frac{\sin t}{t} dt$, which of the following is true ?

- (a) $f(0) > f(1 \cdot 1)$
- (b) $f(0) < f(1 \cdot 1) > f(2 \cdot 1)$
- (c) $f(0) < f(1 \cdot 1) < f(2 \cdot 1) > f(3 \cdot 1)$
- (d) $f(0) < f(1 \cdot 1) < f(2 \cdot 1) < f(3 \cdot 1) > f(4 \cdot 1)$

113. Evaluate : $\int \frac{x^3 + 3x^2 + x + 9}{(x^2 + 1)(x^2 + 3)} dx$.

- (a) $\ln|x^2 + 3| + 3 \tan^{-1} x + c$
- (b) $\frac{1}{2} \ln|x^2 + 3| + \tan^{-1} x + c$
- (c) $\frac{1}{2} \ln|x^2 + 3| + 3 \tan^{-1} x + c$
- (d) $\ln|x^2 + 3| - \tan^{-1} x + c$

114. $\int \frac{\sqrt{\sec^5 x}}{\sqrt{\sin^3 x}} dx$ equals to :

- (a) $(\tan x)^{3/2} - \sqrt{\tan x} + C$
- (b) $2 \left(\frac{1}{3} (\tan x)^{3/2} - \frac{1}{\sqrt{\tan x}} \right) + C$
- (c) $\frac{1}{3} (\tan x)^{3/2} - \sqrt{\tan x} + C$
- (d) $\sqrt{\sin x} + \sqrt{\cos x} + C$

115. $\lim_{x \rightarrow 0} \int_0^x \frac{e^{\sin(tx)}}{x} dt$ equals to :

- (a) 1
- (b) 2
- (c) e
- (d) Does not exist

116. If $A = \int_0^\pi \frac{\sin x}{x^2} dx$, then $\int_0^{\pi/2} \frac{\cos 2x}{x} dx$ is equal to :

- (a) $1 - A$
- (b) $\frac{3}{2} - A$
- (c) $A - 1$
- (d) $1 + A$

Answers

1. (b)	2. (b)	3. (b)	4. (d)	5. (d)	6. (a)	7. (d)	8. (d)	9. (b)	10. (d)
11. (b)	12. (a)	13. (b)	14. (b)	15. (a)	16. (a)	17. (b)	18. (c)	19. (b)	20. (a)
21. (a)	22. (d)	23. (b)	24. (b)	25. (c)	26. (d)	27. (a)	28. (d)	29. (d)	30. (c)
31. (d)	32. (b)	33. (c)	34. (d)	35. (c)	36. (a)	37. (c)	38. (a)	39. (d)	40. (d)
41. (b)	42. (a)	43. (b)	44. (a)	45. (a)	46. (a)	47. (d)	48. (d)	49. (d)	50. (c)
51. (c)	52. (b)	53. (c)	54. (c)	55. (b)	56. (c)	57. (b)	58. (b)	59. (b)	60. (a)
61. (a)	62. (a)	63. (b)	64. (c)	65. (d)	66. (d)	67. (a)	68. (d)	69. (d)	70. (b)
71. (d)	72. (b)	73. (a)	74. (d)	75. (b)	76. (a)	77. (b)	78. (c)	79. (c)	80. (c)
81. (b)	82. (d)	83. (d)	84. (b)	85. (c)	86. (a)	87. (d)	88. (d)	89. (d)	90. (d)
91. (d)	92. (c)	93. (b)	94. (c)	95. (d)	96. (b)	97. (c)	98. (a)	99. (a)	100. (b)
101. (d)	102. (a)	103. (a)	104. (b)	105. (c)	106. (a)	107. (d)	108. (a)	109. (b)	110. (d)
111. (c)	112. (d)	113. (c)	114. (b)	115. (a)	116. (c)				

Exercise-2 : One or More than One Answer is/are Correct

- $\int \frac{dx}{(1+\sqrt{x})^8} = -\frac{1}{3(1+\sqrt{x})^{k_1}} + \frac{2}{7(1+\sqrt{x})^{k_2}} + C$, then :
 (a) $k_1 = 5$ (b) $k_1 = 6$ (c) $k_2 = 7$ (d) $k_2 = 8$
- If $\int_{-\alpha}^{\alpha} (e^x + \cos x \ln(x + \sqrt{1+x^2})) dx > \frac{3}{2}$, then possible value of α can be :
 (a) 1 (b) 2 (c) 3 (d) 4
- For $a > 0$, if $I = \int \sqrt{\frac{x}{a^3 - x^3}} dx = A \sin^{-1} \left(\frac{x^{3/2}}{B} \right) + C$, where C is any arbitrary constant, then :
 (a) $A = \frac{2}{3}$ (b) $B = a^{3/2}$ (c) $A = \frac{1}{3}$ (d) $B = a^{1/2}$
- Let $\int x \sin x \cdot \sec^3 x dx = \frac{1}{2}(x \cdot f(x) - g(x)) + k$, then :
 (a) $f(x) \notin (-1, 1)$ (b) $g(x) = \sin x$ has 6 solution for $x \in [-\pi, 2\pi]$
 (c) $g'(x) = f(x), \forall x \in R$ (d) $f(x) = g(x)$ has no solution
- If $\int (\sin 3\theta + \sin \theta) \cos \theta e^{\sin \theta} d\theta = (A \sin^3 \theta + B \cos^2 \theta + C \sin \theta + D \cos \theta + E) e^{\sin \theta} + F$, then :
 (a) $A = -4$ (b) $B = -12$ (c) $C = -20$ (d) None of these
- For $a > 0$, if $I = \int \sqrt{\frac{x}{a^3 - x^3}} dx = A \sin^{-1} \left(\frac{x^{3/2}}{B} \right) + C$, where C is any arbitrary constant, then :
 (a) $A = \frac{2}{3}$ (b) $B = a^{3/2}$ (c) $A = \frac{1}{3}$ (d) $B = a^{1/2}$
- If $f(\theta) = \lim_{n \rightarrow \infty} \sum_{r=0}^{n\theta} \frac{2r}{n\sqrt{(3\theta n - 2r)(n\theta + 2r)}}$ then :
 (a) $f(1) = \frac{\pi}{6}$ (b) $f(\theta) = \frac{\theta}{2} \int_0^{\theta} \frac{dx}{\sqrt{\theta^2 - \left(x - \frac{\theta}{2}\right)^2}}$
 (c) $f(\theta)$ is a constant function (d) $y = f(\theta)$ is invertible
- If $f(x+y) = f(x)f(y)$ for all x, y and $f(0) \neq 0$, and $F(x) = \frac{f(x)}{1 + (f(x))^2}$ then :
 (a) $\int_{-2010}^{2011} F(x) dx = \int_0^{2011} F(x) dx$ (b) $\int_{-2010}^{2011} F(x) dx - \int_0^{2010} F(x) dx = \int_0^{2011} F(x) dx$
 (c) $\int_{-2010}^{2011} F(x) dx = 0$ (d) $\int_{-2010}^{2010} (2F(-x) - F(x)) dx = 2 \int_0^{2010} F(x) dx$

9. Let $J = \int_{-1}^2 \left(\cot^{-1} \frac{1}{x} + \cot^{-1} x \right) dx$, $K = \int_{-2\pi}^{7\pi} \frac{\sin x}{|\sin x|} dx$. Then which of the following alternative(s) is/are correct?

- (a) $2J + 3K = 8\pi$ (b) $4J^2 + K^2 = 26\pi^2$ (c) $2J - K = 3\pi$ (d) $\frac{J}{K} = \frac{2}{5}$

10. Which of the following function(s) is/are even?

- (a) $f(x) = \int_0^x \ln(t + \sqrt{1+t^2}) dt$ (b) $g(x) = \int_0^x \frac{(2^t + 1)t}{2^t - 1} dt$
 (c) $h(x) = \int_0^x (\sqrt{1+t+t^2} - \sqrt{1-t+t^2}) dt$ (d) $l(x) = \int_0^x \ln\left(\frac{1-t}{1+t}\right) dt$

11. Let $l_1 = \lim_{x \rightarrow \infty} \sqrt{\frac{x - \cos^2 x}{x + \sin x}}$ and $l_2 = \lim_{h \rightarrow 0^+} \int_{-1}^1 \frac{h dx}{h^2 + x^2}$. Then:

- (a) Both l_1 and l_2 are less than $22/7$
 (b) One of the two limits is rational and other irrational
 (c) $l_2 > l_1$
 (d) l_2 is greater than 3 times of l_1

12. For $a > 0$, if $I = \int \sqrt{\frac{x}{a^3 - x^3}} dx = A \sin^{-1} \left(\frac{x^{3/2}}{B} \right) + C$, where C is any arbitrary constant, then:

- (a) $A = \frac{2}{3}$ (b) $B = a^{3/2}$ (c) $A = \frac{1}{3}$ (d) $B = a^{1/2}$

13. If $\int \frac{dx}{1 - \sin^4 x} = a \tan x + b \tan^{-1}(c \tan x) + D$, then:

- (a) $a = \frac{1}{2}$ (b) $b = \sqrt{2}$ (c) $c = \sqrt{2}$ (d) $b = \frac{1}{2\sqrt{2}}$

14. The value of definite integral:

$$\int_{-2014}^{2014} \frac{dx}{1 + \sin^{2015}(x) + \sqrt{1 + \sin^{4030}(x)}} \text{ equals :}$$

- (a) 0 (b) 2014 (c) $(2014)^2$ (d) 4028

15. Let $L = \lim_{n \rightarrow \infty} \int_a^\infty \frac{n dx}{1 + n^2 x^2}$ where $a \in \mathbb{R}$ then L can be:

- (a) π (b) $\frac{\pi}{2}$ (c) 0 (d) $\frac{\pi}{3}$

16. Let $I = \int_0^1 \sqrt{\frac{1+\sqrt{x}}{1-\sqrt{x}}} dx$ and $J = \int_0^1 \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}} dx$ then correct statement(s) is/are :

(a) $I + J = 2$

(b) $I - J = \pi$

(c) $I = \frac{2+\pi}{2}$

(d) $J = \frac{4-\pi}{2}$

Answers

1.	(b, c)	2.	(a, b, c, d)	3.	(a, b)	4.	(a, c, d)	5.	(a, b, c)	6.	(a, b)
7.	(a, b, d)	8.	(b, d)	9.	(a, b)	10.	(a, b, c, d)	11.	(a, b, c, d)	12.	(a, b)
13.	(a, c)	14.	(b)	15.	(a, b, c)	16.	(b, c)				

Exercise-3 : Comprehension Type Problems

Paragraph for Question Nos. 1 to 2

Let $f(x) = \int x^2 \cos^2 x (2x + 6 \tan x - 2x \tan^2 x) dx$ and $f(x)$ passes through the point $(\pi, 0)$

- If $f: R - (2n+1)\frac{\pi}{2} \rightarrow R$ then $f(x)$ be a :
 (a) even function (b) odd function
 (c) neither even nor odd (d) even as well as odd both
- The number of solution(s) of the equation $f(x) = x^3$ in $[0, 2\pi]$ be :
 (a) 0 (b) 3 (c) 2 (d) None of these

Paragraph for Question Nos. 3 to 4

Let $f(x)$ be a twice differentiable function defined on $(-\infty, \infty)$ such that $f(x) = f(2-x)$ and $f'\left(\frac{1}{2}\right) = f'\left(\frac{1}{4}\right) = 0$. Then

- The minimum number of values where $f''(x)$ vanishes on $[0, 2]$ is :
 (a) 2 (b) 3 (c) 4 (d) 5
- $\int_{-1}^1 f'(1+x) x^2 e^{x^2} dx$ is equal to :
 (a) 1 (b) π (c) 2 (d) 0
- $\int_0^1 f(1-t) e^{-\cos \pi t} dt - \int_1^2 f(2-t) e^{\cos \pi t} dt$ is equal to :
 (a) $\int_0^2 f'(t) e^{\cos \pi t} dt$ (b) 1 (c) 2 (d) π

Paragraph for Question Nos. 6 to 8

Consider the function $f(x)$ and $g(x)$, both defined from $R \rightarrow R$

$$f(x) = \frac{x^3}{2} + 1 - x \int_0^x g(t) dt \text{ and } g(x) = x - \int_0^1 f(t) dt, \text{ then}$$

- Minimum value of $f(x)$ is :
 (a) 0 (b) 1 (c) $\frac{3}{2}$ (d) Does not exist

7. The number of points of intersection of $f(x)$ and $g(x)$ is/are :

- (a) 0 (b) 1 (c) 2 (d) 3

8. The area bounded by $g(x)$ with co-ordinate axes is (in square units) :

- (a) $\frac{9}{4}$ (b) $\frac{9}{2}$ (c) $\frac{9}{8}$ (d) None of these

Paragraph for Question Nos. 9 to 11

Let $f(x)$ be function defined on $[0, 1]$ such that $f(1) = 0$ and for any $a \in (0, 1]$,

$$\int_0^a f(x) dx - \int_a^1 f(x) dx = 2f(a) + 3a + b \text{ where } b \text{ is constant.}$$

9. $b =$

- (a) $\frac{3}{2e} - 3$ (b) $\frac{3}{2e} - \frac{3}{2}$ (c) $\frac{3}{2e} + 3$ (d) $\frac{3}{2e} + \frac{3}{2}$

10. The length of the subtangent of the curve $y = f(x)$ at $x = 1/2$ is :

- (a) $\sqrt{e} - 1$ (b) $\frac{\sqrt{e} - 1}{2}$ (c) $\sqrt{e} + 1$ (d) $\frac{\sqrt{e} + 1}{2}$

11. $\int_0^1 f(x) dx =$

- (a) $\frac{1}{e}$ (b) $\frac{1}{2e}$ (c) $\frac{3}{2e}$ (d) $\frac{2}{e}$

Paragraph for Question Nos. 12 to 13

Let $f_0(x) = \ln x$ and for $n \geq 0$ and $x > 0$

Let $f_{n+1}(x) = \int_0^x f_n(t) dt$ then :

12. $f_3(x)$ equals :

- (a) $\frac{x^3}{3} \left(\ln x - \frac{5}{6} \right)$ (b) $\frac{x^3}{3} \left(\ln x - \frac{11}{6} \right)$ (c) $\frac{x^3}{3} \left(\ln x - \frac{11}{6} \right)$ (d) $\frac{x^3}{3} \left(\ln x - \frac{5}{6} \right)$

13. Value of $\lim_{n \rightarrow \infty} \frac{\left(\frac{1}{n} \right) f_n(1)}{\ln(n)}$:

- (a) 0 (b) 1 (c) -1 (d) $-e$

Paragraph for Question Nos. 14 to 15

Let $f: \mathbb{R} \rightarrow \left[\frac{3}{4}, \infty\right)$ be a surjective quadratic function with line of symmetry $2x - 1 = 0$ and $f(1) = 1$

14. If $g(x) = \frac{f(x) + f(-x)}{2}$ then $\int \frac{dx}{\sqrt{g(e^x) - 2}}$ is equal to :

- (a) $\sec^{-1}(e^{-x}) + C$ (b) $\sec^{-1}(e^x) + C$ (c) $\sin^{-1}(e^{-x}) + C$ (d) $\sin^{-1}(e^x) + C$

(Where C is constant of integration)

15. $\int \frac{e^x}{f(e^x)} dx$

(a) $\cot^{-1}\left(\frac{2e^x - 1}{\sqrt{3}}\right) + C$

(b) $\frac{2}{\sqrt{3}} \cot^{-1}\left(\frac{2e^x + 1}{\sqrt{3}}\right) + C$

(c) $\tan^{-1}\left(\frac{2e^x + 1}{\sqrt{3}}\right) + C$

(d) $\frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{2e^x - 1}{\sqrt{3}}\right) + C$

Paragraph for Question Nos. 16 to 17

Let $g(x) = x^C e^{Cx}$ and $f(x) = \int_0^x t e^{2t} (1 + 3t^2)^{1/2} dt$. If $L = \lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)}$ is non-zero finite number then :

16. The value of C is :

- (a) 7 (b) $\frac{3}{2}$ (c) 2 (d) 3

17. The value of L is :

- (a) $\frac{2}{7}$ (b) $\frac{1}{2}$ (c) $\frac{\sqrt{3}}{4}$ (d) $\frac{\sqrt{3}}{2}$

Answers

1. (a)	2. (b)	3. (c)	4. (d)	5. (a)	6. (b)	7. (a)	8. (c)	9. (a)	10. (a)
11. (c)	12. (c)	13. (c)	14. (b)	15. (d)	16. (c)	17. (d)			


Exercise-4 : Matching Type Problems

1.

	Column-I		Column-II
(A)	$\lim_{n \rightarrow \infty} 4 \left[\frac{e^{\frac{1}{n}}}{n^2} + \frac{2}{n^2} e^{\frac{2}{n}} + \frac{3}{n^2} e^{\frac{3}{n}} + \dots + \frac{1}{n} e \right] =$	(P)	0
(B)	$\int_0^1 \ln \left(\frac{1}{x} - 1 \right) dx =$	(Q)	1
(C)	$\int_0^{10\pi} \left(\lim_{x \rightarrow y} \left(\frac{\sin x - \sin y}{x - y} \right) \right) dy =$	(R)	2
(D)	$\int_0^\infty \frac{\ln \left(x + \frac{1}{x} \right) dx}{(1+x^2)} = \frac{\pi}{2} \ln a$, then $a =$	(S)	4
		(T)	5

2. Match the following $\int f(x) dx$ is equal to, if

	Column-I		Column-II
(A)	$f(x) = \frac{1}{(x^2+1)\sqrt{x^2+2}}$	(P)	$\frac{x^5}{5(1-x^4)^{5/2}} + C$
(B)	$f(x) = \frac{1}{(x+2)\sqrt{x^2+6x+7}}$	(Q)	$\sin^{-1} \left(\frac{x+1}{(x+2)\sqrt{2}} \right) + C$
(C)	$f(x) = \frac{x^4+x^8}{(1-x^4)^{7/2}}$	(R)	$(\sqrt{x}-2)\sqrt{1-x} + \cos^{-1} \sqrt{x} + C$
(D)	$f(x) = \frac{\sqrt{1-\sqrt{x}}}{\sqrt{1+\sqrt{x}}}$	(S)	$-\tan^{-1} \sqrt{1+\frac{2}{x^2}} + C$
		(T)	$\frac{x^6}{6(1-x^4)^{5/2}} + C$

3.

Column-I		Column-II	
(A)	$\int_0^{\pi/2} \frac{\cos x}{(1 + \sin x)(2 + \sin x)} dx =$	(P)	$\frac{\pi}{6}$
(B)	$\frac{41\pi}{4} \int_0^4 \cos x dx =$	(Q)	$20 + \frac{1}{\sqrt{2}}$
(C)	$\int_{-1/2}^{1/2} \left([x] + \ln \left(\frac{1+x}{1-x} \right) \right) dx =$ where $[]$ greatest integer function	(R)	$\ln 4 - \ln 3$
(D)	$\int_0^{\pi/2} \frac{2\sqrt{\cos \theta}}{3(\sqrt{\sin \theta} + \sqrt{\cos \theta})} d\theta =$	(S)	$-\frac{1}{2}$

4.

Column-I		Column-II	
(A)	If quadratic equation $3x^2 + ax + 1 = 0$ and $2x^2 + bx + 1 = 0$ have a common root then value of $5ab - 2a^2 - 3b^2 =$	(P)	6
(B)	Number of solution of $x^4 - 2x^2 \sin^2 \frac{\pi x}{2} + 1 = 0$ is/are	(Q)	1
(C)	Number of points of discontinuity $y = \frac{1}{u^2 + u - 2}$ where $u = \frac{1}{x-1}$ is/are	(R)	2
(D)	$\int \frac{dx}{\sqrt[3]{x^{5/2}(1+x)^{7/2}}} = A \left(\frac{x+1}{x} \right)^{-1/A} + C$ (Where C is integration constant), then A =	(S)	3

5. :

		Column-II	
(A)	$\int_0^{1.5} [x^2] dx$	(P)	$-\pi$
(B)	$\int_0^4 \{\sqrt{x}\} dx$ where $\{x\}$ denotes the fractional part of x	(Q)	$4(\sqrt{2} - 1)$
(C)	$\int_0^{2\pi} [\sin x + \cos x] dx$	(R)	$\frac{7}{3}$
(D)	$\int_0^{\pi} \sin x - \cos x dx$	(S)	$2 - \sqrt{2}$

Answers

1. $A \rightarrow S$; $B \rightarrow P$; $C \rightarrow P$; $D \rightarrow S$
2. $A \rightarrow S$; $B \rightarrow Q$; $C \rightarrow P$; $D \rightarrow R$
3. $A \rightarrow R$; $B \rightarrow Q$; $C \rightarrow S$; $D \rightarrow P$
4. $A \rightarrow Q$; $B \rightarrow R$; $C \rightarrow S$; $D \rightarrow P$
5. $A \rightarrow S$; $B \rightarrow R$; $C \rightarrow P$; $D \rightarrow Q$

Exercise-5 : Subjective Type Problems

1. $\int \frac{x + (\arccos 3x)^2}{\sqrt{1-9x^2}} dx = \frac{1}{k_1} \left(\sqrt{1-9x^2} + (\cos^{-1} 3x)^{k_2} \right) + C$, then $k_1^2 + k_2^2 =$
(where C is an arbitrary constant.)
2. If $\int_0^\infty \frac{x^3}{(a^2 + x^2)^5} dx = \frac{1}{ka^6}$, then find the value of $\frac{k}{8}$.
3. Let $f(x) = x \cos x$; $x \in \left[\frac{3\pi}{2}, 2\pi \right]$ and $g(x)$ be its inverse. If $\int_0^{2\pi} g(x) dx = \alpha\pi^2 + \beta\pi + \gamma$, where α, β and $\gamma \in \mathbb{R}$, then find the value of $2(\alpha + \beta + \gamma)$.
4. If $\int (x^6 + x^4 + x^2) \sqrt{2x^4 + 3x^2 + 6} dx = \frac{(\alpha x^6 + \beta x^4 + \gamma x^2)^{3/2}}{18} + C$ where C is constant, then find the value of $(\beta + \gamma - \alpha)$.
5. If the value of the definite integral $\int_{-1}^1 \cot^{-1} \left(\frac{1}{\sqrt{1-x^2}} \right) \cdot \left(\cot^{-1} \frac{x}{\sqrt{1-(x^2)^{|x|}}} \right) dx = \frac{\pi^2(\sqrt{a} - \sqrt{b})}{\sqrt{c}}$
where $a, b, c \in \mathbb{N}$ in their lowest form, then find the value of $(a + b + c)$.
6. The value of $\int \frac{\tan x}{\tan^2 x + \tan x + 1} dx = x - \frac{2}{\sqrt{A}} \tan^{-1} \left(\frac{2 \tan x + 1}{\sqrt{A}} \right) + C$
Then the value of A is :
7. Let $\int_0^1 \frac{4x^3 (1 + (x^4)^{2010})}{(1 + x^4)^{2012}} dx = \frac{\lambda}{\mu}$
where λ and μ are relatively prime positive integers. Find unit digit of μ .
8. Let $\int_1^{\sqrt{3}} \left(x^{2x^2+1} + \ln(x^{2x^{2x^2+1}}) \right) dx = N$. Find the value of $(N - 6)$.
9. If $\int \frac{dx}{\cos^3 x - \sin^3 x} = A \tan^{-1}(f(x)) + B \ln \left| \frac{\sqrt{2} + f(x)}{\sqrt{2} - f(x)} \right| + C$ where $f(x) = \sin x + \cos x$ find the value of $(12A + 9\sqrt{2}B) - 3$.
10. Find the value of $|a|$ for which the area of triangle included between the coordinate axes and any tangent to the curve $x^a y = \lambda^a$ is constant (where λ is constant.)
11. Let $I = \int_0^\pi x^6 (\pi - x)^8 dx$, then $\frac{\pi^{15}}{({}^{15}C_9)I} =$

12. If maximum value of $\int_0^1 (f(x))^3 dx$ under the condition $-1 \leq f(x) \leq 1$; $\int_0^1 f(x) dx = 0$ is $\frac{p}{q}$ (where p and q are relatively prime positive integers.), Find $p + q$.
13. Let a differentiable function $f(x)$ satisfies $f(x) \cdot f'(-x) = f(-x) \cdot f'(x)$ and $f(0) = 1$. Find the value of $\int_{-2}^2 \frac{dx}{1 + f(x)}$.
14. If $\{x\}$ denotes the fractional part of x , then $I = \int_0^{100} \{\sqrt{x}\} dx$, then the value of $\frac{9I}{155}$ is :
15. Let $I_n = \int_0^{\pi} \frac{\sin\left(n + \frac{1}{2}\right)x}{\sin\left(\frac{x}{2}\right)} dx$ where $n \in W$. If $I_1^2 + I_2^2 + I_3^2 + \dots + I_{20}^2 = m\pi^2$, then find the largest prime factor of m .
16. If M be the maximum value of $72 \int_0^y \sqrt{x^4 + (y - y^2)^2} dx$ for $y \in [0, 1]$, then find $\frac{M}{6}$.
17. Find the number of points where $f(\theta) = \int_{-1}^1 \frac{\sin \theta dx}{1 - 2x \cos \theta + x^2}$ is discontinuous where $\theta \in [0, 2\pi]$.
18. Find the value of $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} \left(1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{n}} \right)$.
19. The maximum value of $\int_{-\pi/2}^{3\pi/2} \sin x \cdot f(x) dx$, subject to the condition $|f(x)| \leq 5$ is M , then $\frac{M}{10}$ is equal to :
20. Given a function g , continuous everywhere such that $g(1) = 5$ and $\int_0^1 g(t) dt = 2$. If $f(x) = \frac{1}{2} \int_0^x (x-t)^2 g(t) dt$, then find the value of $f'''(1) + f''(1)$.
21. If $f(n) = \frac{1}{\pi} \int_0^{\pi/2} \frac{\sin^2(n\theta) d\theta}{\sin^2 \theta}$, $n \in N$, then evaluate $\frac{f(15) + f(3)}{f(12) - f(10)}$.
22. Let $f(2-x) = f(2+x)$ and $f(4-x) = f(4+x)$. Function $f(x)$ satisfies $\int_0^2 f(x) dx = 5$. If $\int_0^{50} f(x) dx = I$. Find $[\sqrt{I} - 3]$. (where $[\cdot]$ denotes greatest integer function.)

23. Let $I_n = \int_{-1}^1 |x| \left(1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \dots + \frac{x^{2n}}{2n} \right) dx$. If $\lim_{n \rightarrow \infty} I_n$ can be expressed as rational $\frac{p}{q}$ in its

lowest form, then find the value of $\frac{pq(p+q)}{10}$.

24. Let $\lim_{n \rightarrow \infty} n^{\frac{1}{2} \left(1 + \frac{1}{n} \right)} \cdot (1^1 \cdot 2^2 \cdot 3^3 \cdot \dots \cdot n^n)^{\frac{1}{n^2}} = e^{\frac{-p}{q}}$

where p and q are relative prime positive integers. Find the value of $|p+q|$.

25. If $\int_a^b |\sin x| dx = 8$ and $\int_0^{a+b} |\cos x| dx = 9$ then the value of $\frac{1}{\sqrt{2}x} \left| \int_a^b x \sin x dx \right|$ is :

26. If $f(x)$, $g(x)$, $h(x)$ and $\phi(x)$ are polynomial in x ,

$$\left(\int_1^x f(x) h(x) dx \right) \left(\int_1^x g(x) \phi(x) dx \right) - \left(\int_1^x f(x) \phi(x) dx \right) \left(\int_1^x g(x) h(x) dx \right)$$

is divisible by $(x-1)^\lambda$. Find maximum value of λ .

27. If $\int_0^2 (3x^2 - 3x + 1) \cos(x^3 - 3x^2 + 4x - 2) dx = a \sin(b)$, where a and b are positive integers.

Find the value of $(a+b)$.

28. let $f(x) = \int_0^x e^{x-y} f'(y) dy - (x^2 - x + 1)e^x$

Find the number of roots of the equation $f(x) = 0$.

29. For a positive integer n , let $I_n = \int_{-\pi}^{\pi} \left(\frac{\pi}{2} - |x| \right) \cos nx dx$

Find the value of $[I_1 + I_2 + I_3 + I_4]$ where $[\cdot]$ denotes greatest integer function.

Answers

1.	90	2.	3	3.	3	4.	7	5.	7	6.	3	7.	1
8.	7	9.	8	10.	1	11.	9	12.	5	13.	2	14.	3
15.	5	16.	4	17.	3	18.	2	19.	2	20.	7	21.	9
22.	8	23.	3	24.	5	25.	2	26.	4	27.	2	28.	1
29.	4												

AREA UNDER CURVES

Exercise-1 : Single Choice Problems

- The area enclosed by the curve $[x + 3y] = [x - 2]$ where $x \in [3, 4]$ is :
(where $[\cdot]$ denotes greatest integer function.)
(a) $\frac{2}{3}$ (b) $\frac{1}{3}$ (c) $\frac{1}{4}$ (d) 1
- The area of region enclosed by the curves $y = x^2$ and $y = \sqrt{|x|}$ is :
(a) $\frac{1}{3}$ (b) $\frac{2}{3}$ (c) $\frac{4}{3}$ (d) $\frac{16}{3}$
- Area enclosed by the figure described by the equation $x^4 + 1 = 2x^2 + y^2$, is :
(a) 2 (b) $\frac{16}{3}$ (c) $\frac{8}{3}$ (d) $\frac{4}{3}$
- The area defined by $|y| \leq e^{-|x|} - \frac{1}{2}$ in cartesian co-ordinate system, is :
(a) $(4 - 2\ln 2)$ (b) $(4 - \ln 2)$ (c) $(2 - \ln 2)$ (d) $(2 - 2\ln 2)$
- For each positive integer $n > 1$; A_n represents the area of the region restricted to the following two inequalities : $\frac{x^2}{n^2} + y^2 \leq 1$ and $x^2 + \frac{y^2}{n^2} \leq 1$. Find $\lim_{n \rightarrow \infty} A_n$.
(a) 4 (b) 1 (c) 2 (d) 3
- The ratio in which the area bounded by curves $y^2 = 12x$ and $x^2 = 12y$ is divided by the line $x = 3$ is :
(a) 7 : 15 (b) 15 : 49 (c) 1 : 3 (d) 17 : 49
- The value of positive real parameter 'a' such that area of region bounded by parabolas $y = x - ax^2$, $ay = x^2$ attains its maximum value is equal to :
(a) $\frac{1}{2}$ (b) 2 (c) $\frac{1}{3}$ (d) 1

8. For $0 < r < 1$, let n_r denotes the line that is normal to the curve $y = x^r$ at the point $(1, 1)$. Let S_r denotes the region in the first quadrant bounded by the curve $y = x^r$; the x -axis and the line n_r . Then the value of r that minimizes the area of S_r is :
- (a) $\frac{1}{\sqrt{2}}$ (b) $\sqrt{2} - 1$ (c) $\frac{\sqrt{2}-1}{2}$ (d) $\sqrt{2} - \frac{1}{2}$
9. The area bounded by $|x| = 1 - y^2$ and $|x| + |y| = 1$ is :
- (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{2}{3}$ (d) 1
10. Point A lies on curve $y = e^{-x^2}$ and has the coordinate (x, e^{-x^2}) where $x > 0$. Point B has coordinates $(x, 0)$. If 'O' is the origin, then the maximum area of $\triangle AOB$ is :
- (a) $\frac{1}{\sqrt{8e}}$ (b) $\frac{1}{\sqrt{4e}}$ (c) $\frac{1}{\sqrt{2e}}$ (d) $\frac{1}{\sqrt{e}}$
11. The area enclosed between the curves $y = ax^2$ and $x = ay^2$ ($a > 0$) is 1 sq. unit, then the value of a is :
- (a) $\frac{1}{\sqrt{3}}$ (b) $\frac{1}{2}$ (c) 1 (d) $\frac{1}{3}$
12. Let $f(x) = x^3 - 3x^2 + 3x + 1$ and g be the inverse of it ; then area bounded by the curve $y = g(x)$ with x -axis between $x = 1$ to $x = 2$ is (in square units) :
- (a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) $\frac{3}{4}$ (d) 1
13. Area bounded by $x^2y^2 + y^4 - x^2 - 5y^2 + 4 = 0$ is equal to :
- (a) $\frac{4\pi}{3} + \sqrt{2}$ (b) $\frac{4\pi}{3} - \sqrt{2}$ (c) $\frac{4\pi}{3} + 2\sqrt{3}$ (d) none of these
14. Let $f : R^+ \rightarrow R^+$ is an invertible function such that $f'(x) > 0$ and $f''(x) > 0 \forall x \in [1, 5]$. If $f(1) = 1$ and $f(5) = 5$ and area bounded by $y = f(x)$, x -axis, $x = 1$ and $x = 5$ is 8 sq. units. Then the area bounded by $y = f^{-1}(x)$, x -axis, $x = 1$ and $x = 5$ is :
- (a) 12 (b) 16 (c) 18 (d) 20
15. A circle centered at origin and having radius π units is divided by the curve $y = \sin x$ in two parts. Then area of the upper part equals to :
- (a) $\frac{\pi^2}{2}$ (b) $\frac{\pi^3}{4}$ (c) $\frac{\pi^3}{2}$ (d) $\frac{\pi^3}{8}$
16. The area of the loop formed by $y^2 = x(1 - x^3)$ is :
- (a) $\int_0^1 \sqrt{x - x^4} dx$ (b) $2 \int_0^1 \sqrt{x - x^4} dx$
 (c) $\int_{-1}^1 \sqrt{x - x^4} dx$ (d) $4 \int_0^{1/2} \sqrt{x - x^4} dx$

17. If $f(x) = \min \left[x^2, \sin \frac{x}{2}, (x - 2\pi)^2 \right]$, the area bounded by the curve $y = f(x)$, x -axis, $x = 0$ and $x = 2\pi$ is given by

(Note : x_1 is the point of intersection of the curves x^2 and $\sin \frac{x}{2}$; x_2 is the point of intersection of the curves $\sin \frac{x}{2}$ and $(x - 2\pi)^2$)

- (a) $\int_0^{x_1} \left(\sin \frac{x}{2} \right) dx + \int_{x_1}^{\pi} x^2 dx + \int_{\pi}^{x_2} (x - 2\pi)^2 dx + \int_{x_2}^{2\pi} \left(\sin \frac{x}{2} \right) dx$
- (b) $\int_0^{x_1} x^2 dx + \int_{x_1}^{x_2} \left(\sin \frac{x}{2} \right) dx + \int_{x_2}^{2\pi} (x - 2\pi)^2 dx$, where $x_1 \in \left(0, \frac{\pi}{3} \right)$ and $x_2 \in \left(\frac{5\pi}{3}, 2\pi \right)$
- (c) $\int_0^{x_1} x^2 dx + \int_{x_1}^{x_2} \sin \left(\frac{x}{2} \right) dx + \int_{x_2}^{2\pi} (x - 2\pi)^2 dx$, where $x_1 \in \left(\frac{\pi}{3}, \frac{\pi}{2} \right)$ and $x_2 \in \left(\frac{3\pi}{2}, 2\pi \right)$
- (d) $\int_0^{x_1} x^2 dx + \int_{x_1}^{x_2} \sin \left(\frac{x}{2} \right) dx + \int_{x_2}^{2\pi} (x - 2\pi)^2 dx$, where $x_1 \in \left(\frac{\pi}{2}, \frac{2\pi}{3} \right)$ and $x_2 \in (\pi, 2\pi)$

18. The area enclosed between the curves $|x| + |y| \geq 2$ and $y^2 = 4 \left(1 - \frac{x^2}{9} \right)$ is :

- (a) $(6\pi - 4)$ sq. units (b) $(6\pi - 8)$ sq. units (c) $(3\pi - 4)$ sq. units (d) $(3\pi - 2)$ sq. units

Answers

1. (b)	2. (b)	3. (c)	4. (d)	5. (a)	6. (b)	7. (d)	8. (b)	9. (c)	10. (a)
11. (d)	12. (b)	13. (c)	14. (b)	15. (c)	16. (b)	17. (b)	18. (b)		

Exercise-2 : One or More than One Answer is/are Correct

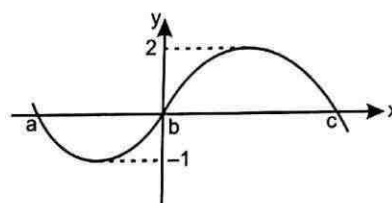
1. Let $f(x)$ be a polynomial function of degree 3 where $a < b < c$ and $f(a) = f(b) = f(c)$. If the graph of $f(x)$ is as shown, which of the following statements are **INCORRECT**? (Where $c > |a|$)

(a) $\int_a^c f(x) dx = \int_b^c f(x) dx + \int_a^b f(x) dx$

(b) $\int_a^c f(x) dx < 0$

(c) $\int_a^b f(x) dx < \int_b^c f(x) dx$

(d) $\frac{1}{b-a} \int_a^b f(x) dx > \frac{1}{c-b} \int_b^c f(x) dx$



2. $T_n = \sum_{r=2n}^{3n-1} \frac{r}{r^2 + n^2}$, $S_n = \sum_{r=2n+1}^{3n} \frac{r}{r^2 + n^2}$, then $\forall n \in \{1, 2, 3, \dots\}$:

(a) $T_n > \frac{1}{2} \ln 2$

(b) $S_n < \frac{1}{2} \ln 2$

(c) $T_n < \frac{1}{2} \ln 2$

(d) $S_n > \frac{1}{2} \ln 2$

3. If a curve $y = a\sqrt{x} + bx$ passes through point $(1, 2)$ and the area bounded by curve, line $x = 4$ and x -axis is 8, then:

(a) $a = 3$

(b) $b = 3$

(c) $a = -1$

(d) $b = -1$

4. Area enclosed by the curves $y = x^2 + 1$ and a normal drawn to it with gradient -1 ; is equal to:

(a) $\frac{2}{3}$

(b) $\frac{1}{3}$

(c) $\frac{3}{4}$

(d) $\frac{4}{3}$

Answers

1.	(b, c, d)	2.	(a, b)	3.	(a, d)	4.	(d)				
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Paragraph for Question Nos. 1 to 3

Let $f: A \rightarrow B$ $f(x) = \frac{x+a}{bx^2+cx+2}$, where A represent domain set and B represent range set of function $f(x)$, $a, b, c \in R$, $f(-1) = 0$ and $y = 1$ is an asymptote of $y = f(x)$ and $y = g(x)$ is the inverse of $f(x)$.

1. $g(0)$ is equal to :
- (a) -1 (b) -3 (c) $-\frac{5}{2}$ (d) $-\frac{3}{2}$
2. Area bounded between the curves $y = f(x)$ and $y = g(x)$ is :
- (a) $2\sqrt{5} + \ln\left(\frac{3-\sqrt{5}}{3+\sqrt{5}}\right)$ (b) $3\sqrt{5} + 2\ln\left(\frac{3+\sqrt{5}}{3-\sqrt{5}}\right)$
- (c) $3\sqrt{5} + 4\ln\left(\frac{3-\sqrt{5}}{3+\sqrt{5}}\right)$ (d) $3\sqrt{5} + 2\ln\left(\frac{3-\sqrt{5}}{3+\sqrt{5}}\right)$
3. Area of region enclosed by asymptotes of curves $y = f(x)$ and $y = g(x)$ is :
- (a) 4 (b) 9 (c) 12 (d) 25

Paragraph for Question Nos. 4 to 6

For $j = 0, 1, 2, \dots, n$ let S_j be the area of region bounded by the x -axis and the curve $ye^x = \sin x$ for $j\pi \leq x \leq (j+1)\pi$

4. The value of S_0 is :
- (a) $\frac{1}{2}(1 + e^\pi)$ (b) $\frac{1}{2}(1 + e^{-\pi})$ (c) $\frac{1}{2}(1 - e^{-\pi})$ (d) $\frac{1}{2}(e^\pi - 1)$
5. The ratio $\frac{S_{2009}}{S_{2010}}$ equals :
- (a) $e^{-\pi}$ (b) e^π (c) $\frac{1}{2}e^\pi$ (d) $2e^\pi$
6. The value of $\sum_{j=0}^{\infty} S_j$ equals to :
- (a) $\frac{e^\pi(1 + e^\pi)}{2(e^\pi - 1)}$ (b) $\frac{1 + e^\pi}{2(e^\pi - 1)}$ (c) $\frac{1 + e^\pi}{e^\pi - 1}$ (d) $\frac{e^\pi(1 + e^\pi)}{(e^\pi - 1)}$

Answers

1.	(a)	2.	(d)	3.	(b)	4.	(b)	5.	(b)	6.	(b)						
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Exercise-4 : Matching Type Problems

1.

Column-I		Column-II	
(A)	Area of region formed by points (x, y) satisfying $[x]^2 = [y]^2$ for $0 \leq x \leq 4$ is equal to (where $[\]$ denotes greatest integer function)	(P)	48
(B)	The area of region formed by points (x, y) satisfying $x + y \leq 6$, $x^2 + y^2 \leq 6y$ and $y^2 \leq 8x$ is $\frac{k\pi - 2}{12}$, then $k =$	(Q)	27
(C)	The area in the first quadrant bounded by the curve $y = \sin x$ and the line $\frac{2y - 1}{\sqrt{2} - 1} = \frac{2}{\pi}(6x - \pi)$ is $\left[\frac{\sqrt{3} - \sqrt{2}}{2} - \frac{(\sqrt{2} + 1)\pi}{k} \right]$, then $k =$	(R)	7
(D)	If the area bounded by the graph of $y = xe^{-ax}$ ($a > 0$) and the abscissa axis is $\frac{1}{9}$ then the value of 'a' is equal to	(S)	4
		(T)	3

Answers

1. A $\rightarrow$ R; B $\rightarrow$ Q; C $\rightarrow$ P; D $\rightarrow$ T

Exercise-5 : Subjective Type Problems

- Let f be a differentiable function satisfying the condition $f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)}$ ($y \neq 0, f(y) \neq 0$) $\forall x, y \in \mathbb{R}$ and $f'(1) = 2$. If the smaller area enclosed by $y = f(x)$, $x^2 + y^2 = 2$ is A , then find $[A]$, where $[]$ represents the greatest integer function.
- Let $f(x)$ be a function which satisfy the equation $f(xy) = f(x) + f(y)$ for all $x > 0, y > 0$ such that $f'(1) = 2$. Let A be the area of the region bounded by the curves $y = f(x)$, $y = |x^3 - 6x^2 + 11x - 6|$ and $x = 0$, then find value of $\frac{28}{17}A$.
- If the area bounded by circle $x^2 + y^2 = 4$, the parabola $y = x^2 + x + 1$ and the curve $y = \left[\sin^2 \frac{x}{4} + \cos \frac{x}{4} \right]$, (where $[]$ denotes the greatest integer function) and x -axis is $\left(\sqrt{3} + \frac{2\pi}{3} - \frac{1}{k} \right)$, then the numerical quantity k should be :
- Let the function $f : [-4, 4] \rightarrow [-1, 1]$ be defined implicitly by the equation $x + 5y - y^5 = 0$. If the area of triangle formed by tangent and normal to $f(x)$ at $x = 0$ and the line $y = 5$ is A , find $\frac{A}{13}$.
- Area of the region bounded by $[x]^2 = [y]^2$, if $x \in [1, 5]$, where $[]$ denotes the greatest integer function, is :
- Consider $y = x^2$ and $f(x)$ where $f(x)$, is a differentiable function satisfying $f(x+1) + f(z-1) = f(x+z) \forall x, z \in \mathbb{R}$ and $f(0) = 0; f'(0) = 4$. If area bounded by curve $y = x^2$ and $y = f(x)$ is Δ , find the value of $\left(\frac{3}{16} \Delta \right)$.
- The least integer which is greater than or equal to the area of region in $x-y$ plane satisfying $x^6 - x^2 + y^2 \leq 0$ is :
- The set of points (x, y) in the plane satisfying $x^{2/5} + |y| = 1$ form a curve enclosing a region of area $\frac{p}{q}$ square units, where p and q are relatively prime positive integers. Find $p - q$.

Answers

1.	1	2.	7	3.	6	4.	5	5.	8	6.	2	7.	2
8.	1												



DIFFERENTIAL EQUATIONS

Exercise-1 : Single Choice Problems

- $\frac{dy}{dx} \left(\frac{1 + \cos x}{y} \right) = -\sin x$ and $f\left(\frac{\pi}{2}\right) = -1$, then $f(0)$ is :
 (a) 2 (b) 1 (c) 3 (d) 4
- The differential equation satisfied by family of curves $y = Ae^x + Be^{3x} + Ce^{5x}$ where A, B, C are arbitrary constants is :
 (a) $\frac{d^3y}{dx^3} - 9\frac{d^2y}{dx^2} + 23\frac{dy}{dx} + 15y = 0$ (b) $\frac{d^3y}{dx^3} + 9\frac{d^2y}{dx^2} - 23\frac{dy}{dx} - 15y = 0$
 (c) $\frac{d^3y}{dx^3} + 9\frac{d^2y}{dx^2} - 23\frac{dy}{dx} + 15y = 0$ (d) $\frac{d^3y}{dx^3} - 9\frac{d^2y}{dx^2} + 23\frac{dy}{dx} - 15y = 0$
- If $y = y(x)$ and it follows the relation $e^{xy^2} + y \cos(x^2) = 5$ then $y'(0)$ is equal to :
 (a) 4 (b) -16 (c) -4 (d) 16
- $(x^2 + y^2)dy = xydx$. If $y(x_0) = e$, $y(1) = 1$, then the value of x_0 is equal to :
 (a) $\sqrt{3}e$ (b) $\sqrt{e^2 - \frac{1}{2}}$ (c) $\sqrt{\frac{e^2 - 1}{2}}$ (d) $\sqrt{e^2 + \frac{1}{2}}$
- The differential equation $\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{y}$ determines a family of circles with :
 (a) Variable radii and fixed centre at $(0, 1)$
 (b) Variable radii and fixed centre at $(0, -1)$
 (c) Fixed radius 1 and variable centres along x-axis
 (d) Fixed radius 1 and variable centres along y-axis
- Interval contained in the domain of definition of non-zero solutions of the differential equation $(x-3)^2y' + y = 0$ is :
 (a) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (b) $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ (c) $\left(\frac{\pi}{8}, \frac{5\pi}{4}\right)$ (d) $(-\pi, \pi)$

7. A function $y = f(x)$ satisfies the differential equation $(x+1)f'(x) - 2(x^2+x)f(x) = \frac{e^{x^2}}{(x+1)}$;
 $\forall x > -1$. If $f(0) = 5$, then $f(x)$ is :
- (a) $\left(\frac{3x+5}{x+1}\right) \cdot e^{x^2}$ (b) $\left(\frac{6x+5}{x+1}\right) \cdot e^{x^2}$
 (c) $\left(\frac{6x+5}{(x+1)^2}\right) \cdot e^{x^2}$ (d) $\left(\frac{5-6x}{x+1}\right) \cdot e^{x^2}$
8. The solution of the differential equation $2x^2y \frac{dy}{dx} = \tan(x^2y^2) - 2xy^2$ given $y(1) = \sqrt{\frac{\pi}{2}}$ is :
- (a) $\sin(x^2y^2) - 1 = 0$ (b) $\cos\left(\frac{\pi}{2} + x^2y^2\right) + x = 0$
 (c) $\sin(x^2y^2) = e^{x-1}$ (d) $\sin(x^2y^2) = e^{2(x-1)}$
9. The differential equation whose general solution is given by $y = C_1 \cos(x+C_2) - C_3 e^{-x+C_4} + C_5 \sin x$, where $C_1, C_2, \dots, C_5$ are constants is :
- (a) $\frac{d^4y}{dx^4} - \frac{d^2y}{dx^2} + y = 0$ (b) $\frac{d^3y}{dx^3} - \frac{d^2y}{dx^2} + \frac{dy}{dx} - y = 0$
 (c) $\frac{d^3y}{dx^3} + \frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 0$ (d) $\frac{d^3y}{dx^3} + \frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$
10. If $y = e^{(\alpha+1)x}$ be solution of differential equation $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$; then α is :
- (a) 0 (b) 1 (c) -1 (d) 2
11. The order and degree of the differential equation $\left(\frac{dy}{dx}\right)^{1/3} - 4\frac{d^2y}{dx^2} - 7x = 0$ are α and β , then the value of $(\alpha + \beta)$ is :
- (a) 3 (b) 4 (c) 2 (d) 5
12. General solution of differential equation of $f(x) \frac{dy}{dx} = f^2(x) + f(x)y + f'(x)y$ is :
 (c being arbitrary constant.)
- (a) $y = f(x) + ce^x$ (b) $y = -f(x) + ce^x$
 (c) $y = -f(x) + ce^x f(x)$ (d) $y = c f(x) + e^x$
13. The order and degree respectively of the differential equation of all tangent lines to parabola $x^2 = 2y$ is :
- (a) 1, 2 (b) 2, 1
 (c) 1, 1 (d) 1, 3

14. The general solution of the differential equation $\frac{dy}{dx} + x(x+y) = x(x+y)^3 - 1$ is :

- (a) $\ln \left| \frac{(x+y+1)(x+y-1)}{(x+y)^4} \right| = x^2 + C$ (b) $\ln \left| \frac{(x+y+1)(x+y-1)}{(x+y)^2} \right| = x^2 + C$
 (c) $2 \ln \left| \frac{(x+y+1)(x+y-1)}{(x+y)^2} \right| = x^2 + C$ (d) $\ln \left| \frac{(x+y+1)(x+y-1)}{(x+y)^2} \right| = x + C$

(where C is arbitrary constant.)

15. The general solution of $\frac{dy}{dx} = 2y \tan x + \tan^2 x$ is :

- (a) $y \cos^2 x = \frac{x}{2} - \frac{\sin 2x}{4} + C$ (b) $y \sec^2 x = \frac{x}{2} - \frac{\sin 2x}{4} + C$
 (c) $y \cos^2 x = \frac{x}{2} - \frac{\cos 2x}{4} + C$ (d) $y \cos^2 x = \frac{x}{2} - \frac{\sin 2x}{2} + C$

(where C is an arbitrary constant.)

16. The solution of differential equation $\frac{d^2y}{dx^2} = \frac{dy}{dx}$, $y(0) = 3$ and $y'(0) = 2$:

- (a) is a periodic function (b) approaches to zero as $x \rightarrow -\infty$
 (c) has an asymptote parallel to x -axis (d) has an asymptote parallel to y -axis

17. The solution of the differential equation $(x^2 + 1) \frac{d^2y}{dx^2} = 2x \left(\frac{dy}{dx} \right)$ under the conditions $y(0) = 1$

and $y'(0) = 3$, is :

- (a) $y = x^2 + 3x + 1$ (b) $y = x^3 + 3x + 1$
 (c) $y = x^4 + 3x + 1$ (d) $y = 3 \tan^{-1} x + x^2 + 1$

18. The differential equation of the family of curves $cy^2 = 2x + c$ (where c is an arbitrary constant.) is :

- (a) $\frac{xdy}{dx} = 1$ (b) $\left(\frac{dy}{dx} \right)^2 = \frac{2xdy}{dx} + 1$ (c) $y^2 = 2xy \frac{dy}{dx} + 1$ (d) $y^2 = \frac{2ydy}{dx} + 1$

19. The solution of the equation $\frac{dy}{dx} + \frac{1}{x} \tan y = \frac{1}{x^2} \tan y \sin y$ is :

- (a) $2y = \sin y (1 - 2cx^2)$ (b) $2x = \cot y (1 + 2cx^2)$
 (c) $2x = \sin y (1 - 2cx^2)$ (d) $2x \sin y = 1 - 2cx^2$

20. Solution of the differential equation $xdy - ydx - \sqrt{x^2 + y^2} dx = 0$ is :

- (a) $y - \sqrt{x^2 + y^2} = cx^2$ (b) $y + \sqrt{x^2 + y^2} = cx$
 (c) $x - \sqrt{x^2 + y^2} = cx^2$ (d) $y + \sqrt{x^2 + y^2} = cx^2$

21. Let $f(x)$ be differentiable function on the interval $(0, \infty)$ such that $f(1) = 1$ and $\lim_{t \rightarrow x} \left(\frac{t^3 f(x) - x^3 f(t)}{t^2 - x^2} \right) = \frac{1}{2} \forall x > 0$, then $f(x)$ is :
- (a) $\frac{1}{4x} + \frac{3x^2}{4}$ (b) $\frac{3}{4x} + \frac{x^3}{4}$ (c) $\frac{1}{4x} + \frac{3x^3}{4}$ (d) $\frac{1}{4x^3} + \frac{3x}{4}$
22. The population $p(t)$ at time ' t ' of a certain mouse species satisfies the differential equation $\frac{d}{dt} p(t) = 0.5 p(t) - 450$. If $p(0) = 850$, then the time at which the population becomes zero is :
- (a) $\frac{1}{2} \ln 18$ (b) $\ln 18$ (c) $2 \ln 18$ (d) $\ln 9$
23. The solution of the differential equation $\sin 2y \frac{dy}{dx} + 2 \tan x \cos^2 y = 2 \sec x \cos^3 y$ is :
(where C is arbitrary constant)
- (a) $\cos y \sec x = \tan x + C$ (b) $\sec y \cos x = \tan x + C$
(c) $\sec y \sec x = \tan x + C$ (d) $\tan y \sec x = \sec x + C$
24. The solution of the differential equation $\frac{dy}{dx} = (4x + y + 1)^2$ is :
(where C is arbitrary constant)
- (a) $4x + y + 1 = 2 \tan(2x + y + C)$ (b) $4x + y + 1 = 2 \tan(x + 2y + C)$
(c) $4x + y + 1 = 2 \tan(2y + C)$ (d) $4x + y + 1 = 2 \tan(2x + C)$
25. If a curve is such that line joining origin to any point $P(x, y)$ on the curve and the line parallel to y -axis through P are equally inclined to tangent to curve at P , then the differential equation of the curve is :
- (a) $x \left(\frac{dy}{dx} \right)^2 - 2y \frac{dy}{dx} = x$ (b) $x \left(\frac{dy}{dx} \right)^2 + 2y \frac{dy}{dx} = x$
(c) $y \left(\frac{dy}{dx} \right)^2 - 2x \frac{dy}{dx} = x$ (d) $y \left(\frac{dy}{dx} \right)^2 - 2y \frac{dy}{dx} = x$
26. If $y = f(x)$ satisfy the differential equation $\frac{dy}{dx} + \frac{y}{x} = x^2$; $f(1) = 1$; then value of $f(3)$ equals :
- (a) 7 (b) 5 (c) 9 (d) 27
27. Let $y = f(x)$ and $\frac{x dy}{y dx} = \frac{3x^2 - y}{2y - x^2}$; $f(1) = 1$ then the possible value of $\frac{1}{3} f(3)$ equals :
- (a) 9 (b) 4 (c) 3 (d) 2

Answers

1. (a)	2. (d)	3. (b)	4. (a)	5. (c)	6. (a)	7. (b)	8. (c)	9. (c)	10. (b)
11. (d)	12. (c)	13. (a)	14. (b)	15. (a)	16. (c)	17. (b)	18. (c)	19. (c)	20. (d)
21. (c)	22. (c)	23. (c)	24. (d)	25. (a)	26. (a)	27. (c)			

Exercise-2 : One or More than One Answer is/are Correct

1. Let $y = f(x)$ be a real valued function satisfying $x \frac{dy}{dx} = x^2 + y - 2$, $f(1) = 1$, then :
- (a) $f(x)$ is minimum at $x = 1$ (b) $f(x)$ is maximum at $x = 1$
 (c) $f(3) = 5$ (d) $f(2) = 3$
2. Solution of differential equation $x \cos x \left(\frac{dy}{dx} \right) + y(x \sin x + \cos x) = 1$ is :
- (a) $xy = \sin x + c \cos x$ (b) $xy \sec x = \tan x + c$
 (c) $xy + \sin x + c \cos x = 0$ (d) None of these
 (where C is an arbitrary constant.)
3. If a differentiable function satisfies $(x - y)f(x + y) - (x + y)f(x - y) = 2(x^2y - y^3) \forall x, y \in \mathbb{R}$ and $f(1) = 2$, then :
- (a) $f(x)$ must be polynomial function (b) $f(3) = 12$
 (c) $f(0) = 0$ (d) $f(3) = 13$
4. A function $y = f(x)$ satisfies the differential equation
 $f(x) \sin 2x - \cos x + (1 + \sin^2 x) f'(x) = 0$
 with $f(0) = 0$. The value of $f\left(\frac{\pi}{6}\right)$ equals to :
- (a) $\frac{2}{5}$ (b) $\frac{3}{5}$ (c) $\frac{1}{5}$ (d) $\frac{4}{5}$
5. Solution of the differential equation $(2 + 2x^2\sqrt{y}) y dx + (x^2\sqrt{y} + 2)x dy = 0$ is/are :
- (a) $xy(x^2\sqrt{y} + 5) = c$ (b) $xy(x^2\sqrt{y} + 3) = c$
 (c) $xy(y^2\sqrt{x} + 3) = c$ (d) $xy(y^2\sqrt{x} + 5) = c$
6. If $y(x)$ satisfies the differential equation $\frac{dy}{dx} = \sin 2x + 3y \cot x$ and $y\left(\frac{\pi}{2}\right) = 2$ then which of the following statement(s) is/are correct ?
- (a) $y\left(\frac{\pi}{6}\right) = 0$ (b) $y'\left(\frac{\pi}{3}\right) = \frac{9 - 3\sqrt{2}}{2}$
 (c) $y(x)$ increases in the interval (d) $\int_{-\pi/2}^{\pi/2} y(x) dx = x$

Answers

1.	(a, c)	2.	(a, b)	3.	(a, b, c)	4.	(a)	5.	(b)	6.	(a, c)
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Exercise-3 : Comprehension Type Problems

Paragraph for Question Nos. 1 to 2

A differentiable function $y = g(x)$ satisfies $\int_0^x (x-t+1)g(t) dt = x^4 + x^2; \forall x \geq 0$.

1. $y = g(x)$ satisfies the differential equation :

(a) $\frac{dy}{dx} - y = 12x^2 + 2$

(b) $\frac{dy}{dx} + 2y = 12x^2 + 2$

(c) $\frac{dy}{dx} + y = 12x^2 + 2$

(d) $\frac{dy}{dx} + y = 12x + 2$

2. The value of $g(0)$ equals to :

(a) 0

(b) 1

(c) e^2

(d) Data insufficient

Paragraph for Question Nos. 3 to 5

Suppose f and g are differentiable functions such that $xg(f(x))f'(g(x))g'(x) = f(g(x))g'(f(x))f'(x) \forall x \in R$ and f is positive, g is positive $\forall x \in R$. Also $\int_0^x f(g(t)) dt = \frac{1}{2}(1 - e^{-2x})$

$\forall x \in R, g(f(0)) = 1$ and $h(x) = \frac{g(f(x))}{f(g(x))} \forall x \in R$.

3. The graph of $y = h(x)$ is symmetric with respect to line :

(a) $x = -1$

(b) $x = 0$

(c) $x = 1$

(d) $x = 2$

4. The value of $f(g(0)) + g(f(0))$ is equal to :

(a) 1

(b) 2

(c) 3

(d) 4

5. The largest possible value of $h(x) \forall x \in R$ is :

(a) 1

(b) $e^{1/3}$

(c) e

(d) e^2

Paragraph for Question Nos. 6 to 8

Given a function ' g ' which has a derivative $g'(x)$ for every real x and which satisfy $g'(0) = 2$ and $g(x+y) = e^y g(x) + e^x g(y)$ for all x and y .

6. The function $g(x)$ is :

(a) $x(2 + xe^x)$

(b) $x(e^x + 1)$

(c) $2xe^x$

(d) $x + \ln(x+1)$

7. The range of function $g(x)$ is :

(a) R

(b) $\left[-\frac{2}{e}, \infty\right)$

(c) $\left[-\frac{1}{e}, \infty\right)$

(d) $[0, \infty)$

8. The value of $\lim_{x \rightarrow -\infty} g(x)$ is :

(a) 0

(b) 1

(c) 2

(d) Does not exist

Answers

1.	(c)	2.	(a)	3.	(c)	4.	(b)	5.	(c)	6.	(c)	7.	(b)	8.	(a)				
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 Exercise-4 : Matching Type Problems

1.

Column-I (Differential equation)		Column-II Solution (Integral curves)	
(A)	$y - x \frac{dy}{dx} = y^2 + \frac{dy}{dx}$	(P)	$y = A_1 x^2 + A_2 x + A_3$
(B)	$(2x - 10y^3) \frac{dy}{dx} + y = 0$	(Q)	$x^2 y^2 + 1 = cy$
(C)	$\left(\frac{dy}{dx}\right)\left(\frac{d^3 y}{dx^3}\right) - 3\left(\frac{d^2 y}{dx^2}\right)^2 = 0$	(R)	$(x+1)(1-y) = cy$
(D)	$(x^2 y^2 - 1) dy + 2xy^3 dx = 0$	(S)	$x = A_1 y^2 + A_2 y + A_3$
		(T)	$xy^2 = 2y^5 + c$

2.

Column-I		Column-II	
(A)	Solution of differential equation $[3x^2 y + 2xy - e^x(1+x)]dx + (x^3 + x^2)dy = 0$ is :	(P)	$y^2(x^2 + 1 + ce^{x^2}) = 1$
(B)	Solution of differential equation $ydx - xdy - 3xy^2 e^{x^2} dx = 0$ is :	(Q)	$(x^2 + x^3)y - xe^x = c$
(C)	Solution of differential equation $\frac{dy}{dx} = xy(x^2 y^2 - 1)$ is :	(R)	$\frac{x}{y} - \frac{3}{2}e^{x^2} = c$
(D)	Solution of differential equation $\frac{dy}{dx}(x^2 y^3 + xy) = 1$ is :	(S)	$\frac{1}{x} = 2 - y^2 + ce^{-y^2/2}$
	(where c is arbitrary constant).	(T)	$\frac{2}{x} = 1 - y^2 + ce^{-y/2}$

 Answers
1. A $\rightarrow$ R; B $\rightarrow$ T; C $\rightarrow$ S; D $\rightarrow$ Q2. A $\rightarrow$ Q; B $\rightarrow$ R; C $\rightarrow$ P; D $\rightarrow$ S

Exercise-5 : Subjective Type Problems

- Find the value of $|a|$ for which the area of triangle included between the coordinate axes and any tangent to the curve $x^a y = \lambda^a$ is constant (where λ is constant.).
- Let $y = f(x)$ satisfies the differential equation $xy(1+y)dx = dy$. If $f(0) = 1$ and $f(2) = \frac{e^2}{k - e^2}$, then find the value of k .
- If $y^2 = 3\cos^2 x + 2\sin^2 x$, then the value of $y^4 + y^3 \frac{d^2 y}{dx^2}$ is
- Let $f(x)$ be a differentiable function in $[-1, \infty)$ and $f(0) = 1$ such that $\lim_{t \rightarrow x+1} \frac{t^2 f(x+1) - (x+1)^2 f(t)}{f(t) - f(x+1)} = 1$. Find the value of $\lim_{x \rightarrow 1} \frac{\ln(f(x)) - \ln 2}{x - 1}$.
- Let $y = (a \sin x + (b+c) \cos x) e^{x+d}$, where a, b, c and d are parameters represent a family of curves, then differential equation for the given family of curves is given by $y'' - \alpha y' + \beta y = 0$, then $\alpha + \beta =$
- Let $y = f(x)$ satisfies the differential equation $xy(1+y)dx = dy$. If $f(0) = 1$ and $f(2) = \frac{e^2}{k - e^2}$, then find the value of k .

Answers

1.	1	2.	2	3.	6	4.	1	5.	4	6.	2		
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Algebra

- 8.** Quadratic Equations
- 9.** Sequence and Series
- 10.** Determinants
- 11.** Complex Numbers
- 12.** Matrices
- 13.** Permutation and Combinations
- 14.** Binomial Theorem
- 15.** Probability
- 16.** Logarithms

QUADRATIC EQUATIONS

Exercise-1 : Single Choice Problems

- Sum of values of x and y satisfying the equation $3^x - 4^y = 77$; $3^{x/2} - 2^y = 7$ is :
 (a) 2 (b) 3 (c) 4 (d) 5
- If $f(x) = \prod_{i=1}^3 (x - a_i) + \sum_{i=1}^3 a_i - 3x$ where $a_i < a_{i+1}$ for $i = 1, 2$, then $f(x) = 0$ has :
 (a) only one distinct real root (b) exactly two distinct real roots
 (c) exactly 3 distinct real roots (d) 3 equal real roots
- Complete set of real values of ' a ' for which the equation $x^4 - 2ax^2 + x + a^2 - a = 0$ has all its roots real :
 (a) $\left[\frac{3}{4}, \infty\right)$ (b) $[1, \infty)$ (c) $[2, \infty)$ (d) $[0, \infty)$
- The cubic polynomial with leading coefficient unity all whose roots are 3 units less than the roots of the equation $x^3 - 3x^2 - 4x + 12 = 0$ is denoted as $f(x)$, then $f'(x)$ is equal to:
 (a) $3x^2 - 12x + 5$ (b) $3x^2 + 12x + 5$ (c) $3x^2 + 12x - 5$ (d) $3x^2 - 12x - 5$
- The set of values of k ($k \in \mathbb{R}$) for which the equation $x^2 - 4|x| + 3 - |k - 1| = 0$ will have exactly four real roots, is :
 (a) $(-2, 4)$ (b) $(-4, 4)$ (c) $(-4, 2)$ (d) $(-1, 0)$
- The number of integers satisfying the inequality $\frac{x}{x+6} \leq \frac{1}{x}$ is :
 (a) 7 (b) 8 (c) 9 (d) 3
- The product of uncommon real roots of the two polynomials $p(x) = x^4 + 2x^3 - 8x^2 - 6x + 15$ and $q(x) = x^3 + 4x^2 - x - 10$ is:
 (a) 4 (b) 6 (c) 8 (d) 12
- If λ_1, λ_2 ($\lambda_1 > \lambda_2$) are two values of λ for which the expression $f(x, y) = x^2 + \lambda xy + y^2 - 5x - 7y + 6$ can be resolved as a product of two linear factors, then the value of $3\lambda_1 + 2\lambda_2$ is :
 (a) 5 (b) 10 (c) 15 (d) 20

9. Let α, β be the roots of the quadratic equation $ax^2 + bx + c = 0$, then the roots of the equation $a(x+1)^2 + b(x+1)(x-2) + c(x-2)^2 = 0$ are :
- (a) $\frac{2\alpha+1}{\alpha-1}, \frac{2\beta+1}{\beta-1}$ (b) $\frac{2\alpha-1}{\alpha+1}, \frac{2\beta-1}{\beta+1}$
 (c) $\frac{\alpha+1}{\alpha-2}, \frac{\beta+1}{\beta-2}$ (d) $\frac{2\alpha+3}{\alpha-1}, \frac{2\beta+3}{\beta-1}$
10. If $a, b \in \mathbb{R}$ distinct numbers satisfying $|a-1| + |b-1| = |a| + |b| = |a+1| + |b+1|$, then the minimum value of $|a-b|$ is :
 (a) 3 (b) 0 (c) 1 (d) 2
11. The smallest positive integer p for which expression $x^2 - 2px + 3p + 4$ is negative for atleast one real x is :
 (a) 3 (b) 4 (c) 5 (d) 6
12. For $x \in \mathbb{R}$, the expression $\frac{x^2 + 2x + c}{x^2 + 4x + 3c}$ can take all real values if $c \in$:
 (a) (1, 2) (b) [0, 1]
 (c) (0, 1) (d) (-1, 0)
13. If 2 lies between the roots of the equation $t^2 - mt + 2 = 0$, ($m \in \mathbb{R}$) then the value of $\left[\left(\frac{3|x|}{9+x^2} \right)^m \right]$ is :
 (where $[\cdot]$ denotes greatest integer function)
 (a) 0 (b) 1 (c) 8 (d) 27
14. The number of integral roots of the equation $x^8 - 24x^7 - 18x^5 + 39x^2 + 1155 = 0$ is :
 (a) 0 (b) 2 (c) 4 (d) 6
15. If the value of $m^4 + \frac{1}{m^4} = 119$, then the value of $\left| m^3 - \frac{1}{m^3} \right| =$
 (a) 11 (b) 18 (c) 24 (d) 36
16. If the equation $ax^2 + 2bx + c = 0$ and $ax^2 + 2cx + b = 0$, $a \neq 0$, $b \neq c$, have a common root, then their other roots are the roots of the quadratic equation:
 (a) $a^2x(x+1) + 4bc = 0$ (b) $a^2x(x+1) + 8bc = 0$
 (c) $a^2x(x+2) + 8bc = 0$ (d) $a^2x(1+2x) + 8bc = 0$
17. If $\cos \alpha, \cos \beta$ and $\cos \gamma$ are the roots of the equation $9x^3 - 9x^2 - x + 1 = 0$; $\alpha, \beta, \gamma \in [0, \pi]$ then the radius of the circle whose centre is $(\Sigma \alpha, \Sigma \cos \alpha)$ and passing through $(2 \sin^{-1}(\tan \pi/4), 4)$ is:
 (a) 2 (b) 3 (c) 4 (d) 5
18. For real values of x , the value of expression $\frac{11x^2 - 12x - 6}{x^2 + 4x + 2}$:

- (a) lies between -17 and -3
 (c) lies between 3 and 17
 (b) does not lie between -17 and -3
 (d) does not lie between 3 and 17
19. $\frac{x+3}{x^2-x-2} \geq \frac{1}{x-4}$ holds for all x satisfying:
 (a) $-2 < x < 1$ or $x > 4$
 (b) $-1 < x < 2$ or $x > 4$
 (c) $x < -1$ or $2 < x < 4$
 (d) $x > -1$ or $2 < x < 4$
20. If $x = 4 + 3i$ (where $i = \sqrt{-1}$), then the value of $x^3 - 4x^2 - 7x + 12$ equals:
 (a) -88
 (b) $48 + 36i$
 (c) $-256 + 12i$
 (d) -84
21. Let $f(x) = \frac{x^2 + x - 1}{x^2 - x + 1}$, then the largest value of $f(x) \forall x \in [-1, 3]$ is:
 (a) $\frac{3}{5}$
 (b) $\frac{5}{3}$
 (c) 1
 (d) $\frac{4}{3}$
22. In above problem, the range of $f(x) \forall x \in [-1, 1]$ is:
 (a) $\left[-1, \frac{3}{5}\right]$
 (b) $\left[-1, \frac{5}{3}\right]$
 (c) $\left[-\frac{1}{3}, 1\right]$
 (d) $[-1, 1]$
23. If the roots of the equation $\frac{1}{x+p} + \frac{1}{x+q} = \frac{1}{r}$ are equal in magnitude but opposite in sign, then the product of the roots is:
 (a) $-2(p^2 + q^2)$
 (b) $-(p^2 + q^2)$
 (c) $-\frac{(p^2 + q^2)}{2}$
 (d) $-pq$
24. If a root of the equation $a_1x^2 + b_1x + c_1 = 0$ is the reciprocal of a root of the equation $a_2x^2 + b_2x + c_2 = 0$, then :
 (a) $(a_1a_2 - c_1c_2)^2 = (a_1b_2 - b_1c_2)(a_2b_1 - b_2c_1)$
 (b) $(a_1a_2 - b_1b_2)^2 = (a_1b_2 - b_1c_2)(a_2b_1 - b_2c_1)$
 (c) $(b_1c_2 - b_2c_1)^2 = (a_1b_2 - b_1c_2)(a_2b_1 + b_2c_1)$
 (d) $(b_1c_2 - b_2c_1)^2 = (a_1b_2 + b_1c_2)(a_2b_1 - b_2c_1)$
25. If $\alpha \neq \beta$ but $\alpha^2 = 5\alpha - 3$ and $\beta^2 = 5\beta - 3$, then the equation with roots $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$ is:
 (a) $3x^2 - 25x + 3 = 0$
 (b) $x^2 + 5x - 3 = 0$
 (c) $x^2 - 5x + 3 = 0$
 (d) $3x^2 - 19x + 3 = 0$
26. If the difference between the roots of $x^2 + ax + b = 0$ is same as that of $x^2 + bx + a = 0, a \neq b$, then:
 (a) $a + b + 4 = 0$
 (b) $a + b - 4 = 0$
 (c) $a - b - 4 = 0$
 (d) $a - b + 4 = 0$
27. If $\tan \theta_i; i = 1, 2, 3, 4$ are the roots of equation $x^4 - x^3 \sin 2\beta + x^2 \cos 2\beta - x \cos \beta - \sin \beta = 0$, then $\tan(\theta_1 + \theta_2 + \theta_3 + \theta_4) =$
 (a) $\sin \beta$
 (b) $\cos \beta$
 (c) $\tan \beta$
 (d) $\cot \beta$

28. Let a, b, c, d are positive real numbers such that $\frac{a}{b} \neq \frac{c}{d}$, then the roots of the equation: $(a^2 + b^2)x^2 + 2x(ac + bd) + (c^2 + d^2) = 0$ are:
- (a) real and distinct (b) real and equal
(c) imaginary (d) nothing can be said
29. If α, β are the roots of $ax^2 + bx + c = 0$, then the equation whose roots are $2 + \alpha, 2 + \beta$, is:
- (a) $ax^2 + x(4a - b) + 4a - 2b + c = 0$ (b) $ax^2 + x(4a - b) + 4a + 2b + c = 0$
(c) $ax^2 + x(b - 4a) + 4a + 2b + c = 0$ (d) $ax^2 + x(b - 4a) + 4a - 2b + c = 0$
30. Minimum possible number of positive root of the quadratic equation $x^2 - (1 + \lambda)x + \lambda - 2 = 0, \lambda \in R$:
- (a) 2 (b) 0
(c) 1 (d) can not be determined
31. Let α, β be real roots of the quadratic equation $x^2 + kx + (k^2 + 2k - 4) = 0$, then the minimum value of $\alpha^2 + \beta^2$ is equal to:
- (a) 12 (b) $\frac{4}{9}$ (c) $\frac{16}{9}$ (d) $\frac{8}{9}$
32. Polynomial $P(x) = x^2 - ax + 5$ and $Q(x) = 2x^3 + 5x - (a - 3)$ when divided by $x - 2$ have same remainders, then 'a' is equal to:
- (a) 10 (b) -10 (c) 20 (d) -20
33. If a and b are non-zero distinct roots of $x^2 + ax + b = 0$, then the least value of $x^2 + ax + b$ is equal to:
- (a) $\frac{2}{3}$ (b) $\frac{9}{4}$ (c) $-\frac{9}{4}$ (d) 1
34. Let α, β be the roots of the equation $ax^2 + bx + c = 0$. A root of the equation $a^3x^2 + abcx + c^3 = 0$ is:
- (a) $\alpha + \beta$ (b) $\alpha^2 + \beta$ (c) $\alpha^2 - \beta$ (d) $\alpha^2\beta$
35. Let a, b, c be the lengths of the sides of a triangle (no two of them are equal) and $k \in R$. If the roots of the equation $x^2 + 2(a + b + c)x + 6k(ab + bc + ca) = 0$ are real, then:
- (a) $k < \frac{2}{3}$ (b) $k > \frac{2}{3}$ (c) $k > 1$ (d) $k < \frac{1}{4}$
36. Root(s) of the equation $9x^2 - 18|x| + 5 = 0$ belonging to the domain of definition of the function $f(x) = \log(x^2 - x - 2)$ is/are:
- (a) $\frac{-5}{3}, \frac{-1}{3}$ (b) $\frac{5}{3}, \frac{1}{3}$ (c) $\frac{-5}{3}$ (d) $\frac{-1}{3}$
37. If $\beta + \cos^2 \alpha, \beta + \sin^2 \alpha$ are the roots of $x^2 + 2bx + c = 0$ and $\gamma + \cos^4 \alpha, \gamma + \sin^4 \alpha$ are the roots of $x^2 + 2Bx + C = 0$, then:
- (a) $b - B = c - C$ (b) $b^2 - B^2 = c - C$ (c) $b^2 - B^2 = 4(c - C)$ (d) $4(b^2 - B^2) = c - C$

38. Minimum value of $|x - p| + |x - 15| + |x - p - 15|$. If $p \leq x \leq 15$ and $0 < p < 15$:
 (a) 30 (b) 15 (c) 10 (d) 0
39. If the quadratic equation $4x^2 - 2x - m = 0$ and $4p(q - r)x^2 - 2q(r - p)x + r(p - q) = 0$ have a common root such that second equation has equal roots then the value of m will be :
 (a) 0 (b) 1 (c) 2 (d) 3
40. The range of k for which the inequality $k \cos^2 x - k \cos x + 1 \geq 0 \forall x \in (-\infty, \infty)$ is :
 (a) $k > -\frac{1}{2}$ (b) $k > 4$ (c) $-\frac{1}{2} \leq k \leq 4$ (d) $\frac{1}{2} \leq k \leq 5$
41. If $\frac{1+\alpha}{1-\alpha}, \frac{1+\beta}{1-\beta}, \frac{1+\gamma}{1-\gamma}$ are roots of the cubic equation $f(x) = 0$ where α, β, γ are the roots of the cubic equation $3x^3 - 2x + 5 = 0$, then the number of negative real roots of the equation $f(x) = 0$ is:
 (a) 0 (b) 1 (c) 2 (d) 3
42. The sum of all integral values of λ for which $(\lambda^2 + \lambda - 2)x^2 + (\lambda + 2)x < 1 \forall x \in R$, is :
 (a) -1 (b) -3 (c) 0 (d) -2
43. If $\alpha, \beta, \gamma, \delta \in R$ satisfy $\frac{(\alpha+1)^2 + (\beta+1)^2 + (\gamma+1)^2 + (\delta+1)^2}{\alpha + \beta + \gamma + \delta} = 4$
 If biquadratic equation $a_0x^4 + a_1x^3 + a_2x^2 + a_3x + a_4 = 0$ has the roots $\left(\alpha + \frac{1}{\beta} - 1\right), \left(\beta + \frac{1}{\gamma} - 1\right), \left(\gamma + \frac{1}{\delta} - 1\right), \left(\delta + \frac{1}{\alpha} - 1\right)$. Then the value of a_2/a_0 is :
 (a) 4 (b) -4 (c) 6 (d) none of these
44. If the complete set of value of x satisfying $|x - 1| + |x - 2| + |x - 3| \geq 6$ is $(-\infty, a] \cup [b, \infty)$, then $a + b =$:
 (a) 2 (b) 3 (c) 6 (d) 4
45. If exactly one root of the quadratic equation $x^2 - (a + 1)x + 2a = 0$ lies in the interval $(0, 3)$, then the set of value 'a' is given by:
 (a) $(-\infty, 0) \cup (6, \infty)$ (b) $(-\infty, 0] \cup (6, \infty)$
 (c) $(-\infty, 0] \cup [6, \infty)$ (d) $(0, 6)$
46. The condition that the root of $x^3 + 3px^2 + 3qx + r = 0$ are in H.P. is :
 (a) $2p^3 - 3pqr + r^2 = 0$ (b) $3p^3 - 2pqr + p^2 = 0$
 (c) $2q^3 - 3pqr + r^2 = 0$ (d) $r^3 - 3pqr + 2q^3 = 0$
47. If x is real and $4y^2 + 4xy + x + 6 = 0$, then the complete set of values of x for which y is real, is:
 (a) $x \leq -2$ or $x \geq 3$ (b) $x \leq 2$ or $x \geq 3$ (c) $x \leq -3$ or $x \geq 2$ (d) $-3 \leq x \leq 2$
48. The solution of the equation $\log_{\cos x^2} (3 - 2x) < \log_{\cos x^2} (2x - 1)$ is :
 (a) $(1/2, 1)$ (b) $(-\infty, 1)$
 (c) $(1/2, 3)$ (d) $(1, \infty) - \sqrt{2n\pi}, n \in N$

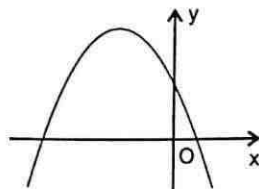
49. If the roots α, β of the equation $px^2 + qx + r = 0$ are real and of opposite sign (where p, q, r are real coefficient), then the roots of the equation $\alpha(x - \beta)^2 + \beta(x - \alpha)^2 = 0$ are:
 (a) positive (b) negative
 (c) real and of opposite sign (d) imaginary
50. Let a, b and c be three distinct real roots of the cubic $x^3 + 2x^2 - 4x - 4 = 0$.
 If the equation $x^3 + qx^2 + rx + s = 0$ has roots $\frac{1}{a}, \frac{1}{b}$ and $\frac{1}{c}$, then the value of $(q + r + s)$ is equal to :
 (a) $\frac{3}{4}$ (b) $\frac{1}{2}$ (c) $\frac{1}{4}$ (d) $\frac{1}{6}$
51. Solution set of the inequality, $2 - \log_2(x^2 + 3x) \geq 0$ is :
 (a) $[-4, 1]$ (b) $[-4, -3] \cup (0, 1]$ (c) $(-\infty, -3) \cup (1, \infty)$ (d) $(-\infty, -4) \cup [1, \infty)$
52. For what least integral 'k' is the quadratic trinomial $(k - 2)x^2 + 8x + (k + 4)$ is positive for all real values of x ?
 (a) $k = 4$ (b) $k = 5$ (c) $k = 3$ (d) $k = 6$
53. If roots of the equation $(m - 2)x^2 - (8 - 2m)x - (8 - 3m) = 0$ are opposite in sign, then number of integral values(s) of m is/are :
 (a) 0 (b) 1 (c) 2 (d) more than 2
54. If $\log_{0.6} \left(\log_6 \left(\frac{x^2 + x}{x + 4} \right) \right) < 0$, then complete set of value of 'x' is :
 (a) $(-4, -3) \cup (8, \infty)$ (b) $(-\infty, -3) \cup (8, \infty)$
 (c) $(8, \infty)$ (d) None of these
55. Two different real numbers α and β are the roots of the quadratic equation $ax^2 + c = 0$ with $a, c \neq 0$, then $\alpha^3 + \beta^3$ is :
 (a) a (b) $-c$ (c) 0 (d) -1
56. The least integral value of 'k' for which $(k - 1)x^2 - (k + 1)x + (k + 1)$ is positive for all real value of x is:
 (a) 1 (b) 2 (c) 3 (d) 4
57. If $(-2, 7)$ is the highest point on the graph of $y = -2x^2 - 4ax + k$, then k equals:
 (a) 31 (b) 11 (c) -1 (d) $-1/3$
58. If $a + b + c = 0$, $a, b, c \in Q$ then roots of the equation $(b + c - a)x^2 + (c + a - b)x + (a + b - c) = 0$ are :
 (a) rational (b) irrational (c) imaginary (d) none of these
59. If two roots of $x^3 - ax^2 + bx - c = 0$ are equal in magnitude but opposite in sign. Then:
 (a) $a + bc = 0$ (b) $a^2 = bc$
 (c) $ab = c$ (d) $a - b + c = 0$

60. If α and β are the real roots of $x^2 + px + q = 0$ and α^4, β^4 are the roots of $x^2 - rx + s = 0$. Then the equation $x^2 - 4qx + 2q^2 - r = 0$ has always ($\alpha \neq \beta, p \neq 0, p, q, r, s \in R$):
 (a) one positive and one negative root (b) two positive roots
 (c) two negative roots (d) can't say anything
61. If $x^2 + px + 1$ is a factor of $ax^3 + bx + c$, then:
 (a) $a^2 + c^2 = -ab$ (b) $a^2 + c^2 = ab$ (c) $a^2 - c^2 = ab$ (d) $a^2 - c^2 = -ab$
62. In a $\triangle ABC$ $\tan \frac{A}{2}, \tan \frac{B}{2}, \tan \frac{C}{2}$ are in H.P., then the value of $\cot \frac{A}{2} \cot \frac{C}{2}$ is :
 (a) 3 (b) 2 (c) 1 (d) $\sqrt{3}$
63. Let $f(x) = 10 - |x - 10| \forall x \in [-9, 9]$, if M and m be the maximum and minimum value of $f(x)$ respectively, then :
 (a) $M + m = 0$ (b) $2M + m = -9$ (c) $2M + m = 7$ (d) $M + m = 7$
64. Solution of the quadratic equation $(3|x| - 3)^2 = |x| + 7$, which belongs to the domain of the function $y = \sqrt{(x - 4)x}$ is :
 (a) $\pm \frac{1}{9}, \pm 2$ (b) $\frac{1}{9}, 8$ (c) $-2, -\frac{1}{9}$ (d) $-\frac{1}{9}, 8$
65. Number of real solutions of the equation $x^2 + 3|x| + 2 = 0$ is :
 (a) 0 (b) 1 (c) 2 (d) 4
66. If the roots of equation $x^2 - bx + c = 0$ be two consecutive integers, then $b^2 - 4c =$
 (a) 3 (b) -2 (c) 1 (d) 2
67. If x is real, then maximum value of $\frac{3x^2 + 9x + 17}{3x^2 + 9x + 7}$ is :
 (a) 41 (b) 1 (c) $\frac{17}{7}$ (d) $\frac{1}{4}$
68. If $\frac{x^2 + 2x + 7}{2x + 3} < 6, x \in R$ then :
 (a) $x \in \left(-\infty, -\frac{3}{2}\right) \cup (11, \infty)$ (b) $x \in (-\infty, -1) \cup (11, \infty)$
 (c) $x \in \left(-\frac{3}{2}, -1\right)$ (d) $x \in \left(-\infty, -\frac{3}{2}\right) \cup (-1, 11)$
69. If x is real, then range of $\frac{3x - 2}{7x + 5}$ is :
 (a) $R - \left\{\frac{2}{5}\right\}$ (b) $R - \left\{\frac{3}{7}\right\}$ (c) $(-\infty, \infty)$ (d) $R - \left\{\frac{-2}{5}\right\}$
70. Let A denotes the set of values of x for which $\frac{x + 2}{x - 4} \leq 0$ and B denotes the set of values of x for which $x^2 - ax - 4 \leq 0$. If B is the subset of A , then a **CAN NOT** take the integral value :
 (a) 0 (b) 1 (c) 2 (d) 3

71. If the quadratic polynomial $P(x) = (p-3)x^2 - 2px + 3p - 6$ ranges from $[0, \infty)$ for every $x \in \mathbb{R}$, then the value of p can be :

- (a) 3 (b) 4 (c) 6 (d) 7

72. If graph of the quadratic $y = ax^2 + bx + c$ is given below :



then :

- (a) $a < 0, b > 0, c > 0$ (b) $a < 0, b > 0, c < 0$
 (c) $a < 0, b < 0, c > 0$ (d) $a < 0, b < 0, c < 0$

73. If quadratic equation $ax^2 + bx + c = 0$ does not have real roots, then which of the following may be false :

- (a) $a(a-b+c) > 0$ (b) $c(a-b+c) > 0$
 (c) $b(a-b+c) > 0$ (d) $(a+b+c)(a-b+c) > 0$

74. Minimum value of $y = x^2 - 3x + 5, x \in [-4, 1]$ is :

- (a) 3 (b) $\frac{11}{4}$ (c) 0 (d) 9

75. If $3x^2 - 17x + 10 = 0$ and $x^2 - 5x + m = 0$ has a common root, then sum of all possible real values of 'm' is :

- (a) 0 (b) $-\frac{26}{9}$ (c) $\frac{29}{9}$ (d) $\frac{26}{3}$

76. For real numbers x and y , if $x^2 + xy - y^2 + 2x - y + 1 = 0$, then :

- (a) y can not be between 0 and $\frac{8}{5}$ (b) y can not be between $-\frac{8}{5}$ and $\frac{8}{5}$
 (c) y can not be between $-\frac{8}{5}$ and 0 (d) y can not be between $-\frac{16}{5}$ and 0

77. If $3x^4 - 6x^3 + kx^2 - 8x - 12$ is divisible by $x - 3$, then it is also divisible by :

- (a) $3x^2 - 4$ (b) $3x^2 + 4$
 (c) $3x^2 + x$ (d) $3x^2 - x$

78. The complete set of values of a so that equation $\sin^4 x + a \sin^2 x + 4 = 0$ has at least one real root is :

- (a) $(\infty, -5]$ (b) $(-\infty, 4] \cup [4, \infty)$
 (c) $(-\infty, -4]$ (d) $[4, \infty)$

79. Let r, s, t be the roots of the equation $x^3 + ax^2 + bx + c = 0$, such that $(rs)^2 + (st)^2 + (rt)^2 = b^2 - kac$, then $k =$

- (a) 1 (b) 2 (c) 3 (d) 4

80. If the roots of the cubic $x^3 + ax^2 + bx + c = 0$ are three consecutive positive integers, then the value of $\frac{a^2}{b+1} =$
- (a) 1 (b) 2 (c) 3 (d) 4
81. Let 'k' be a real number. The minimum number of distinct real roots possible of the equation $(3x^2 + kx + 3)(x^2 + kx - 1) = 0$ is :
- (a) 0 (b) 2 (c) 3 (d) 4
82. If r and s are variables satisfying the equation $\frac{1}{r+s} = \frac{1}{r} + \frac{1}{s}$. The value of $\left(\frac{r}{s}\right)^3$ is equal to :
- (a) 1 (b) -1
(c) 3 (d) not possible to determine
83. Let $f(x) = x^2 + ax + b$. If the maximum and the minimum values of $f(x)$ are 3 and 2 respectively for $0 \leq x \leq 2$, then the possible ordered pair of (a, b) is :
- (a) $(-2, 3)$ (b) $(-3/2, 2)$ (c) $(-5/2, 3)$ (d) $(-5/2, 2)$
84. The roots of the equation $|x^2 - x - 6| = x + 2$ are given by :
- (a) -2, 2, 4 (b) 0, 1, 4 (c) -2, 1, 4 (d) 0, 2, 4
85. If a, b, c be the sides of $\triangle ABC$ and equations $ax^2 + bx + c = 0$ and $5x^2 + 12x + 13 = 0$ have a common root, then $\angle C$ is :
- (a) 60° (b) 90° (c) 120° (d) 45°
86. If α, β and γ are three real roots of the equation $x^3 - 6x^2 + 5x - 1 = 0$, then the value of $\alpha^4 + \beta^4 + \gamma^4$ is :
- (a) 250 (b) 650 (c) 150 (d) 950
87. If one of the roots of the equation $2x^2 - 6x + k = 0$ is $\frac{\alpha + 5i}{2}$, then the value of α and k are :
- (a) $\alpha = 3, k = 8$ (b) $\alpha = \frac{3}{2}, k = 17$ (c) $\alpha = -3, k = -17$ (d) $\alpha = 3, k = 17$
88. Let x_1 and x_2 be the real roots of the equation $x^2 - (k-2)x + (k^2 + 3k + 5) = 0$, then the maximum value of $x_1^2 + x_2^2$ is :
- (a) 19 (b) 18 (c) $\frac{50}{9}$ (d) non-existent
89. The complete set of values of 'a' for which the inequality $(a-1)x^2 - (a+1)x + (a-1) \geq 0$ is true for all $x \geq 2$.
- (a) $\left[\frac{3}{7}, 1\right]$ (b) $(-\infty, 1)$ (c) $\left(-\infty, \frac{7}{3}\right]$ (d) $\left[\frac{7}{3}, \infty\right)$
90. If α, β be the roots of $4x^2 - 17x + \lambda = 0, \lambda \in R$ such that $1 < \alpha < 2$ and $2 < \beta < 3$, then the number of integral values of λ is :
- (a) 1 (b) 2 (c) 3 (d) 4

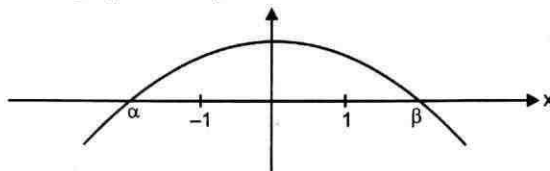
91. Assume that p is a real number. In order of $\sqrt[3]{x+3p+1} - \sqrt[3]{x} = 1$ to have real solutions, it is necessary that:
 (a) $p \geq 1/4$ (b) $p \geq -1/4$ (c) $p \geq 1/3$ (d) $p \geq -1/3$
92. If α, β are the roots of the quadratic equation $x^2 - \left(3 + 2^{\sqrt{\log_2 3}} - 3^{\sqrt{\log_3 2}}\right)x - 2(3^{\log_3 2} - 2^{\log_2 3}) = 0$, then the value of $\alpha^2 + \alpha\beta + \beta^2$ is equal to :
 (a) 3 (b) 5 (c) 7 (d) 11
93. The minimum value of $f(x, y) = x^2 - 4x + y^2 + 6y$ when x and y are subjected to the restrictions $0 \leq x \leq 1$ and $0 \leq y \leq 1$, is :
 (a) -1 (b) -2 (c) -3 (d) -5
94. The expression $ax^2 + 2bx + c$, where ' a ' is non-zero real number, has same sign as that of ' a ' for every real value of x , then roots of quadratic equation $ax^2 + (b-c)x - 2b - c - a = 0$, are :
 (a) real and equal (b) real and unequal
 (c) non-real having positive real part (d) non-real having negative real part
95. Let a, b and c be the roots of $x^3 - x + 1 = 0$, then the value of $\left(\frac{1}{a+1} + \frac{1}{b+1} + \frac{1}{c+1}\right)$ equals to :
 (a) 1 (b) -1 (c) 2 (d) -2
96. The number of integral values of k for which the inequality $x^2 - 2(4k-1)x + 15k^2 - 2k - 7 \geq 0$ holds for all $x \in R$ is :
 (a) 2 (b) 3 (c) 4 (d) infinite
97. The number of integral values which can be taken by the expression, $f(x) = \frac{x^3 - 1}{(x-1)(x^2 - x + 1)}$ for $x \in R$, is :
 (a) 1 (b) 2 (c) 3 (d) infinite
98. The complete set of values of m for which the inequality $\frac{x^2 - mx - 2}{x^2 + mx + 4} > -1$ is satisfied $\forall x \in R$, is :
 (a) $m = 0$ (b) $-1 < m < 1$ (c) $-2 < m < 2$ (d) $-4 < m < 4$
99. The complete set of values of a for which the roots of the equation $x^2 - 2|a+1|x + 1 = 0$ are real is given by :
 (a) $(-\infty, -2] \cup [0, \infty)$ (b) $(-\infty, -1] \cup [0, \infty)$
 (c) $(-\infty, -1] \cup [1, \infty)$ (d) $(-\infty, -2] \cup [1, \infty)$
100. The quadratic polynomials defined on real coefficients $P(x) = a_1x^2 + 2b_1x + c_1$, $Q(x) = a_2x^2 + 2b_2x + c_2$, $P(x)$ and $Q(x)$ both take positive values $\forall x \in R$. If $f(x) = a_1a_2x^2 + b_1b_2x + c_1c_2$, then :
 (a) $f(x) < 0 \forall x \in R$
 (b) $f(x) > 0 \forall x \in R$

- (c) $f(x)$ takes both positive and negative values
 (d) Nothing can be said about $f(x)$
101. If the equation $x^2 + 4 + 3\cos(ax + b) = 2x$ has a solution then a possible value of $(a + b)$ equals
 (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{2}$ (d) π
102. Let α, β be the roots of $x^2 - 4x + A = 0$ and γ, δ be the roots of $x^2 - 36x + B = 0$. If $\alpha, \beta, \gamma, \delta$ form an increasing G.P. and $A^t = B$ then the value of ' t ' equals
 (a) 4 (b) 5 (c) 6 (d) 8
103. How many roots does the following equation possess $3^{|x|} (|2 - |x||) = 1$?
 (a) 2 (b) 3 (c) 4 (d) 6
104. If $\cot \alpha$ equals the integral solution of inequality $4x^2 - 16x + 15 < 0$ and $\sin \beta$ equals to the slope of the bisector of the first quadrant, then $\sin(\alpha + \beta) \sin(\alpha - \beta)$ is equal to :
 (a) $-\frac{3}{5}$ (b) $-\frac{4}{5}$ (c) $\frac{2}{\sqrt{2}}$ (d) 3
105. Consider the functions $f_1(x) = x$ and $f_2(x) = 2 + \log_e x$, $x > 0$, where e is the base of natural logarithm. The graphs of the functions intersect :
 (a) once in $(0, 1)$ and never in $(1, \infty)$ (b) once in $(0, 1)$ and once in (e^2, ∞)
 (c) once in $(0, 1)$ and once in (e, e^2) (d) more than twice in $(0, \infty)$
106. The sum of all the real roots of equation $x^4 - 3x^3 - 2x^2 - 3x + 1 = 0$ is :
 (a) 1 (b) 2 (c) 3 (d) 4
107. If α, β ($\alpha < \beta$) are the real roots of the equation $x^2 - (k + 4)x + k^2 - 12 = 0$ such that $4 \in (\alpha, \beta)$; then the number of integral values of k equal to :
 (a) 4 (b) 5 (c) 6 (d) 7
108. Let α, β be real roots of the quadratic equation $x^2 + kx + (k^2 + 2k - 4) = 0$, then the maximum value of $(\alpha^2 + \beta^2)$ is equal to :
 (a) 9 (b) 10 (c) 11 (d) 12
109. Let $f(x) = a^x - x \ln a$, $a > 1$. Then the complete set of real values of x for which $f'(x) > 0$ is :
 (a) $(1, \infty)$ (b) $(-1, \infty)$ (c) $(0, \infty)$ (d) $(0, 1)$
110. If a, b and c are the roots of the equation $x^3 + 2x^2 + 1 = 0$, find $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$:
 (a) 8 (b) -8 (c) 0 (d) 2
111. Let α, β are the two real roots of equation $x^2 + px + q = 0$, $p, q \in R$, $q \neq 0$. If the quadratic equation $g(x) = 0$ has two roots $\alpha + \frac{1}{\alpha}, \beta + \frac{1}{\beta}$ such that sum of roots is equal to product of roots, then the complete range of q is :

- (a) $\left[\frac{1}{3}, 3\right]$ (b) $\left(\frac{1}{3}, 3\right]$ (c) $\left[\frac{1}{3}, 3\right)$ (d) $\left(-\infty, \frac{1}{3}\right) \cup (3, \infty)$
112. If the equation $\ln(x^2 + 5x) - \ln(x + a + 3) = 0$ has exactly one solution for x , then number of integers in the range of a is :
 (a) 4 (b) 5 (c) 6 (d) 7
113. Let $f(x) = x^2 + \frac{1}{x^2} - 6x - \frac{6}{x} + 2$, then minimum value of $f(x)$ is :
 (a) -2 (b) -8 (c) -9 (d) -12
114. If $x^2 + bx + b$ is a factor of $x^3 + 2x^2 + 2x + c$ ($c \neq 0$), then $b - c$ is :
 (a) 2 (b) -1 (c) 0 (d) -2
115. If roots of $x^3 + 2x^2 + 1 = 0$ are α, β and γ , then the value of $(\alpha\beta)^3 + (\beta\gamma)^3 + (\alpha\gamma)^3$, is :
 (a) -11 (b) 3 (c) 0 (d) -2
116. How many roots does the following equation possess $3^{|x|}(|2 - |x||) = 1$?
 (a) 2 (b) 3 (c) 4 (d) 6
117. The sum of all the real roots of equation $x^4 - 3x^3 - 2x^2 - 3x + 1 = 0$ is :
 (a) 1 (b) 2 (c) 3 (d) 4
118. If α and β are the roots of the quadratic equation $4x^2 + 2x - 1 = 0$ then the value of $\sum_{r=1}^{\infty} (\alpha^r + \beta^r)$ is :
 (a) 2 (b) 3 (c) 6 (d) 0
119. The number of value(s) of x satisfying the equation $(2011)^x + (2012)^x + (2013)^x - (2014)^x = 0$ is/are :
 (a) exactly 2 (b) exactly 1 (c) more than one (d) 0
120. If α, β ($\alpha < \beta$) are the real roots of the equation $x^2 - (k + 4)x + k^2 - 12 = 0$ such that $4 \in (\alpha, \beta)$; then the number of integral values of k equals to :
 (a) 4 (b) 5 (c) 6 (d) 7
121. Let α, β be real roots of the quadratic equation $x^2 + kx + (k^2 + 2k - 4) = 0$, then the maximum value of $(\alpha^2 + \beta^2)$ is equal to :
 (a) 9 (b) 10 (c) 11 (d) 12
122. The exhaustive set of values of a for which inequation $(a - 1)x^2 - (a + 1)x + a - 1 \geq 0$ is true $\forall x \geq 2$
 (a) $(-\infty, 1)$ (b) $\left[\frac{7}{3}, \infty\right)$ (c) $\left[\frac{3}{7}, \infty\right)$ (d) None of these
123. If the equation $x^2 + ax + 12 = 0$, $x^2 + bx + 15 = 0$ and $x^2 + (a + b)x + 36 = 0$ have a common positive root, then $b - 2a$ is equal to.
 (a) -6 (b) 22 (c) 6 (d) -22

124. The equation $e^{\sin x} - e^{-\sin x} - 4 = 0$ has
 (a) infinite number of real roots (b) no real root
 (c) exactly one real root (d) exactly four real roots
125. The difference between the maximum and minimum value of the function $f(x) = 3\sin^4 x - \cos^6 x$ is :
 (a) $\frac{3}{2}$ (b) $\frac{5}{2}$ (c) 3 (d) 4
126. If α, β are the roots of $x^2 - 3x + \lambda = 0$ ($\lambda \in R$) and $\alpha < 1 < \beta$, then the true set of values of λ equals :
 (a) $\lambda \in \left(2, \frac{9}{4}\right]$ (b) $\lambda \in \left(-\infty, \frac{9}{4}\right]$ (c) $\lambda \in (2, \infty)$ (d) $\lambda \in (-\infty, 2)$
127. If $2x^2 + 5x + 7 = 0$ and $ax^2 + bx + c = 0$ have at least one root common such that $a, b, c \in \{1, 2, \dots, 100\}$, then the difference between the maximum and minimum values of $a + b + c$ is :
 (a) 196 (b) 284 (c) 182 (d) 126
128. Two particles, A and B, are in motion in the xy -plane. Their co-ordinates at each instant of time t ($t \geq 0$) are given by $x_A = t, y_A = 2t, x_B = 1 - t$ and $y_B = t$. The minimum distance between particles A and B is :
 (a) $\frac{1}{5}$ (b) $\frac{1}{\sqrt{5}}$ (c) 1 (d) $\sqrt{\frac{2}{3}}$
129. If $a \neq 0$ and the equation $ax^2 + bx + c = 0$ has two roots α and β such that $\alpha < -3$ and $\beta > 2$, which of the following is always true ?
 (a) $a(a + |b| + c) > 0$ (b) $a(a + |b| + c) < 0$
 (c) $9a - 3b + c > 0$ (d) $(9a - 3b + c)(4a + 2b + c) < 0$
130. The number of negative real roots of the equation $(x^2 + 5x)^2 - 24 = 2(x^2 + 5x)$ is :
 (a) 4 (b) 3 (c) 2 (d) 1
131. The number of real values of x satisfying the equation $3|x - 2| + |1 - 5x| + 4|3x + 1| = 13$ is :
 (a) 1 (b) 4 (c) 2 (d) 3
132. If $\log_{\cos x} \sin x \geq 2$ and $0 \leq x \leq 3\pi$ then $\sin x$ lies in the interval
 (a) $\left[\frac{\sqrt{5}-1}{2}, 1\right]$ (b) $\left[0, \frac{\sqrt{5}-1}{2}\right]$ (c) $\left[\frac{1}{2}, 1\right]$ (d) none of these
133. Let $f(x) = x^2 + bx + c$, minimum value of $f(x)$ is -5 , then absolute value of the difference of the roots of $f(x)$ is :
 (a) 5 (b) $\sqrt{20}$
 (c) $\sqrt{15}$ (d) Can't be determined
134. Sum of all the solutions of the equation $|x - 3| + |x + 5| = 7x$, is :
 (a) $\frac{6}{7}$ (b) $\frac{8}{7}$ (c) $\frac{58}{63}$ (d) $\frac{8}{45}$

135. Let $f(x) = x^2 + \frac{1}{x^2} - 6x - \frac{6}{x} + 2$, then minimum value of $f(x)$ is :
 (a) -2 (b) -8 (c) -9 (d) -12
136. If $a + b + c = 1$, $a^2 + b^2 + c^2 = 9$ and $a^3 + b^3 + c^3 = 1$, then the value of $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$ is :
 (a) $\frac{2}{3}$ (b) 5 (c) 6 (d) 1
137. If roots of $x^3 + 2x^2 + 1 = 0$ are α, β and γ , then the value of $(\alpha\beta)^3 + (\beta\gamma)^3 + (\alpha\gamma)^3$, is :
 (a) -11 (b) 3 (c) 0 (d) -2
138. If $x^2 + bx + b$ is a factor of $x^3 + 2x^2 + 2x + c$ ($c \neq 0$), then $b - c$ is :
 (a) 2 (b) -1 (c) 0 (d) -2
139. The graph of quadratic polynomial $f(x) = ax^2 + bx + c$ is shown below



- (a) $\frac{c}{a}|\beta - \alpha| < -2$ (b) $f(x) > 0 \forall x > \beta$ (c) $ac > 0$ (d) $\frac{c}{a} > -1$
140. If $f(x) = \frac{x^2 - 3x + 4}{x^2 + 3x + 4}$, then complete solution of $0 < f(x) < 1$, is :
 (a) $(-\infty, \infty)$ (b) $(0, \infty)$ (c) $(-\infty, 0)$ (d) $(0, 1) \cup (2, \infty)$
141. If α, β, γ are the roots of the equation $x^3 + 2x^2 - x + 1 = 0$, then value of $\frac{(2-\alpha)(2-\beta)(2-\gamma)}{(2+\alpha)(2+\beta)(2+\gamma)}$, is:
 (a) 5 (b) -5 (c) 10 (d) $\frac{5}{3}$
142. If α and β are roots of the quadratic equation $x^2 + 4x + 3 = 0$, then the equation whose roots are $2\alpha + \beta$ and $\alpha + 2\beta$ is :
 (a) $x^2 - 12x + 35 = 0$ (b) $x^2 + 12x - 33 = 0$ (c) $x^2 - 12x - 33 = 0$ (d) $x^2 + 12x + 35 = 0$
143. If a, b, c are real distinct numbers such that $a^3 + b^3 + c^3 = 3abc$, then the quadratic equation $ax^2 + bx + c = 0$ has
 (a) Real roots (b) At least one negative root
 (c) Both roots are negative (d) Non real roots
144. If the equation $x^2 + ax + 12 = 0$, $x^2 + bx + 15 = 0$ and $x^2 + (a+b)x + 36 = 0$ have a common positive root, then $b - 2a$ is equal to.
 (a) -6 (b) 22 (c) 6 (d) -22